Memo version 1.1

BESIII Analysis Memo

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BESIII Analysis Memo (August 13, 2026)

Cross section measurement of process $e^+e^- \to \pi^+\pi^-\pi^0$ from threshold to 2.0 GeV via ISR technique

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Abstract

Using an e^+e^- collision sample of 20 fb^{-1} collected by the BESIII detector at $\sqrt{s}$ = 3.773 GeV under BOSS-7.1.2, we measure the $e^+e^- \to \pi^+\pi^-\pi^0$ cross section via initial-state

radiation (ISR). By tagging an ISR photon, the effective center-of-mass energy is reduced, providing a continuous 3π mass spectrum from threshold to the nominal energy. After unfolding detector effects, the four independent data samples (Round 03/04, 15, 16, and 17) are combined using the Best Linear Unbiased Estimator (BLUE) method, and the dressed cross sections are obtained.

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List of changes

For an easy reading of the document, major changes in various versions with respect to the previous one are listed below.

Version 1.0

- First version of memo.

Version 1.1

- Corrected the NLO fraction calculation.
- Studied the purity of LO/NLO/NLOSA events from the same-sign selection; the effect is canceled by the efficiency matrix.
- Added illustrative figures showing diagonal systematic uncertainties per bin; the full covariance matrices are used in the final results.
- Added the description of the helix parameter correction in the kinematic fit, which was previously omitted.

Contents

1 Introduction

The anomalous magnetic moment of the muon, $a_\mu = (g-2)/2$, is a cornerstone quantity for testing the Standard Model (SM) and probing new physics. In 2025, the Muon $g-2$ Experiment at Fermilab released its final measurement [1, 2] based on the complete Run 1–6 dataset, yielding a world average experimental value combining with the BNL E821 result [3] of

$$a_\mu^{\text{exp}} = 116\,592\,072(1.5) \times 10^{-11} \quad \text{(124 ppb)}.$$

Within the Standard Model, a_μ^{SM} is decomposed into several contributions:

$$a_\mu^{\text{SM}} = a_\mu^{\text{QED}} + a_\mu^{\text{EW}} + a_\mu^{\text{HVP}} + a_\mu^{\text{HLbL}},$$

where a_μ^{QED} denotes the quantum electrodynamics contribution (computed to five loops), a_μ^{EW} the electroweak contribution, a_μ^{HVP} the hadronic vacuum polarization, and a_μ^{HLbL} the hadronic light-by-light scattering. Among these, the hadronic contributions — particularly a_μ^{HVP} — dominate the theoretical uncertainty.

The largest uncertainty in the SM prediction comes from the leading-order hadronic vacuum polarization (LO-HVP) contribution, a_μ^{HVP}, which cannot be computed perturbatively. In the traditional data-driven dispersive approach, a_μ^{HVP} is obtained via the dispersion integral [4, 5]

$$a_\mu^{\text{HVP}} = \frac{\alpha^2}{3\pi^2} \int_{4m_\pi^2}^{\infty} \frac{ds}{s} K(s)\, R(s),$$

where $R(s) = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-)$ is the hadronic R-ratio, and $K(s)$ is a known QED kernel function that peaks at low energies. Consequently, high-precision measurements of exclusive hadronic cross sections at e^+e^- colliders are indispensable for a reliable SM prediction.

The WP2020 [6] evaluation by the Muon $g-2$ Theory Initiative relied on this data-driven method. It yielded a SM prediction of

$$a_\mu^{\text{SM}}(\text{WP2020}) = 116\,591\,810(43) \times 10^{-11},$$

which, when compared with the then-available experimental value, resulted in a $> 5\sigma$ deviation, fueling intense interest in possible new physics.

However, the theoretical landscape has undergone a fundamental shift with the publication of the 2025 White Paper (WP2025) [7]. Persistent tensions among different $e^+e^- \to \pi^+\pi^-$ measurements — in particular the higher $\pi\pi$ cross section reported by the CMD-3 experiment [8] compared to other earlier experiments — have made it impossible to produce a meaningful combined data-driven average. In contrast, lattice QCD [9] calculations of the LO-HVP have reached sub-percent precision and show good internal consistency among different collaborations.

Consequently, WP2025 adopts a consolidated lattice-QCD average as its central value for the LO-HVP contribution, rather than the data-driven result used in WP2020. This change shifts the total SM prediction upward to

$$a_\mu^{\text{SM}}(\text{WP2025}) = 116\,592\,033(62) \times 10^{-11}.$$

The comparison between theory and experiment now yields:

$$a_\mu^{\mathrm{exp}} - a_\mu^{\mathrm{SM}}(\mathrm{WP2025}) = 38(63) \times 10^{-11},$$

corresponding to a negligible deviation of approximately 0.6σ. The long-standing $> 5\sigma$ discrepancy has effectively been resolved with the updated SM prediction. Nevertheless, this resolution does not imply that the problem is fully understood. The disagreement between the data-driven and lattice-QCD evaluations of the LO-HVP term remains unresolved, as does the tension between CMD-3 and other $e^+e^- \rightarrow \pi^+\pi^-$ experiments. Resolving these discrepancies requires new, independent, high-precision measurements of the key hadronic cross sections, particularly those with the largest contributions to a_μ^{HVP}.

For the 3π channel, the Belle II experiment reported a new measurement in 2024 [10]. However, concerns have been raised regarding the reliability of its uncertainty estimation, leading some recent SM theory compilations to explicitly exclude Belle II data from their averages[11]. This situation highlights the urgent need for independent cross section measurements from other facilities.

The BESIII experiment, with its large data samples, is ideally suited to provide a reliable and independent measurement of the $e^+e^- \rightarrow \pi^+\pi^-\pi^0$ cross section in the low-energy region critical for a_μ^{HVP}. In this memo, we use an e^+e^- collision sample corresponding to an integrated luminosity of 20 fb^{-1} collected at a center-of-mass energy of 3.773 GeV. We present a measurement based on the initial-state radiation (ISR) technique using the process $e^+e^- \rightarrow \pi^+\pi^-\pi^0\gamma_{\mathrm{ISR}}$, which allows access to the 3π hadronic cross section from continuum production without energy scanning. This work aims to provide a valid and systematically well-controlled measurement, thereby adding reliable experimental input for the 3π channel and contributing to the ongoing effort to reconcile the data-driven and lattice-QCD approaches.

2 BEPCII and BESIII detector

The Beijing Electron-Positron Collider II (BEPCII) —a double-ring multi-bunch collider —is operating in the τ-charm energy region (1.84 to 4.95 GeV) with the design luminosity of 1×10^{33} $\mathrm{cm}^{-2}\mathrm{s}^{-1}$ [13]. The Beijing Spectrometer III (BESIII) operating at BEPCII is a high-precision, general-purpose detector that records symmetric e^+e^- collisions provided by the BEPCII storage ring. BESIII is approximately cylindrically symmetric with 93% coverage of the solid angle around the e^+e^- interaction point (IP). The components of the apparatus ordered by distance from the IP, are a 43-layer small-cell main drift chamber (MDC), a time-of-flight (TOF) system based on plastic scintillators with two layers in the barrel region and one layer in the end-cap region (the end-cap TOF system has been updated with the multi-gap resistive plate chamber in 2014 [12]), a 6240-cell CsI(Tl) crystal electromagnetic calorimeter (EMC), a superconducting solenoid magnet providing a 1.0 T magnetic field aligned with the beam axis, and resistive plate muon-counter layers interleaved with steel. The momentum resolution for charged tracks in the MDC is 0.5% for transverse momenta with 1 GeV/c. The energy resolution in the EMC is 2.5% in the barrel region and 5.0% in the end-cap region for 1 GeV photons. For charged tracks, particle identification (PID) for charged tracks combines measurements of the energy deposit dE/dx in MDC and flight time in TOF and forms likelihoods $\mathcal{L}(h)(h = K, \pi)$ for a hadron h hypothesis. More details about the BESIII detector are provided in Ref [14].

Table 1: Beam energies and integrated luminosities of the data samples used in this work, quoted from Ref. [15, 16].

E_{beam}(GeV)	Integrated luminosity (pb^{-1})	Data taking time	RunNo
3.773	2931.8 ± 0.2 ± 13.8 [15]	2010(round03)	11414 – 13988, 14395 – 14604
		2011(round04)	20448 – 23454
3.773	4995.0 ± 19.0 [16]	2022(round15)	70522 – 73929
3.773	8157.0 ± 31.0 [16]	2023(round16)	74031 – 78536
3.773	4191.0 ± 16.0 [16]	2024(round17)	78615 – 81094

Table 2: Processes and luminosity scales of the inclusive MC samples used in this analysis. Separate samples are available for each data-taking round. Most processes use samples with an integrated luminosity equivalent to ten times that of the data.

Process	Luminosity scale
$e^+e^- \to \psi(3770) \to D^0\bar{D}^0$	$10\times$
$e^+e^- \to \psi(3770) \to D^+D^-$	$10\times$
$e^+e^- \to \tau^+\tau^-$	$10\times$
$e^+e^- \to \psi(3770) \to$ non-$D\bar{D}$	$10\times$
$e^+e^- \to q\bar{q}$	$10\times$
$e^+e^- \to \gamma_{\mathrm{ISR}}\psi(2S)$	$10\times$
$e^+e^- \to \gamma_{\mathrm{ISR}}J/\psi$	$10\times$
$e^+e^- \to e^+e^-$	$0.25\times$
$e^+e^- \to \gamma\gamma$	$3\times$
$e^+e^- \to \mu^+\mu^-$	$15\times$

3 Data and MC samples

We use an e^+e^- collision sample corresponding to an integrated luminosity of 20 fb^{-1}, collected with the BESIII detector at a center-of-mass energy of 3.773 GeV, reconstucted under BOSS-7.1.2 version. The details are listed in Table 1, quoted from Refs. [15, 16].

Simulated Monte Carlo (MC) samples are employed to study backgrounds, optimize event selection, and estimate detection efficiencies. Two categories of MC samples are utilized in this analysis:

- **The inclusive MC samples** comprehensively model known physics processes at $\sqrt{s}$ = 3.773 GeV, including $D\bar{D}$ production, $\tau^+\tau^-$ pair production, initial-state-radiation (ISR) production of vector charmonium states (J/ψ and $\psi(2S)$), QED processes, and continuum light hadron production. The charm working group provides separate samples for the different data-taking rounds(round03-04, round15, round16, and round17) [18]. In this analysis, we use inclusive MC samples with an integrated luminosity equivalent to ten times that of the data for most processes, except for the QED processes $e^+e^- \to e^+e^-$, $\gamma\gamma$, and $\mu^+\mu^-$, where we use the samples with the scaling factors provided by the charm working group. The processes and the corresponding luminosity scales of the samples used in this analysis are summarized in Table 2.

- **Exclusive signal and background MC samples** generated specifically for this analysis using the PHOKHARA event generator [19, 20, 21] (version 9.1). These samples are produced independently for each data-taking round (round03, round04, round15, round16, round17) to account for variations in detector conditions. Sufficient statistics are generated for each process to ensure reliable efficiency determination and background studies. The generated processes include:
  - $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$ (signal process)
  - $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$
  - $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\rm ISR}$

  These exclusive samples are used for signal efficiency studies, background characterization, and validation of the analysis methodology.

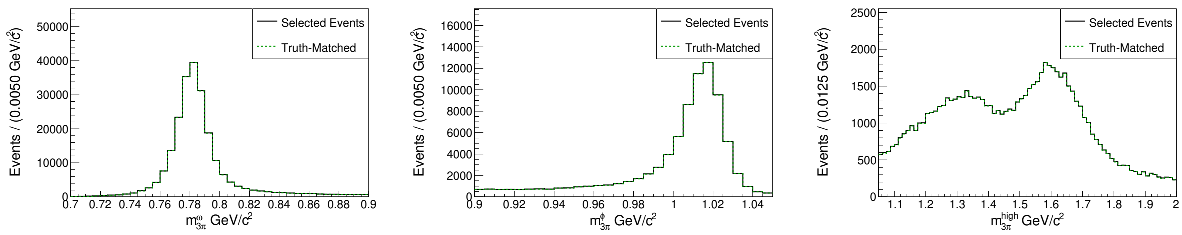
4 Event Selections

The process under investigation is $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$, whose final states include two charged tracks, one neutral π^0, and one ISR photon. The following selection criteria are applied to all the data samples. Here the figures are shown by taking $\sqrt{s} = 3.773$ GeV round16 data as an example. Situations are similar to each other.

- Good charged track: A charged track is taken as a good MDC track if it originates from the beam vertex and lies within the MDC polar angle region:
  - $|\cos\theta| < 0.93$, where θ is the polar angle with respect to the beam axis.
  - $|\Delta r| < 1$ cm, $|\Delta z| < 10$ cm, where $|\Delta r|$ and $|\Delta z|$ is distance of closest approach to the interaction point (IP) perpendicular to the beam axis and along the beam axis, respectively.
  - the ratio of the calorimeter-deposited energy to the track momentum E_{EMC}/p is required to be less than 0.8 for both of the charged tracks.
  - no PID requirements for π^+, π^-.
- γ selection:
  - $E > 0.025$ GeV (barrel: $|\cos\theta| < 0.8$), $E > 0.050$ GeV (endcap: $0.86 < |\cos\theta| < 0.92$).
  - $0 \leqslant t_{\rm TDC} \leqslant 14(\times 50\text{ns})$ to suppress fake photons due to electronic noise or beam background.
  - the closest angle with all charged tracks $\theta > 10^\circ$ to suppress background from radiation of charged tracks.
- $\pi^0 \to \gamma\gamma$ selection:
  - two photon invariant mass $M_{\gamma\gamma}$ within (0.115, 0.150) GeV/c^2.
  - To improve the momentum resolution, 1C kinematic fit is performed. Here, the 1C means π^0 mass constraint, and the updated 4-momentum will be used in the further analysis. $\chi^2_{\rm KF} < 10$

Candidate events are selected by requiring exactly two good charged tracks, accompanied by at least one reconstructed π^0 and a minimum of three good photon showers. This ensures the presence of at least one π^0 and one isolated γ_{ISR}. A four-constraint (4C) kinematic fit is then applied to the event, and the candidate with the smallest χ^2_{4C} value is chosen for further analysis.

After identifying an ISR photon from the isolated photon showers, the invariant mass of the remaining $e^+e^- \to \pi^+\pi^-\pi^0$ system is reconstructed. As expected for ISR processes, the resulting center-of-mass energy $\sqrt{s'}$ of the $\pi^+\pi^-\pi^0$ system shows a continuous spectrum, characteristic of the radiative return mechanism. This continuum behavior is illustrated in Fig. 1 across the $\omega(782)$, $\phi(1020)$, and ω'/ω'' mass regions.



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω' and ω'' region

Figure 1: Invariant mass spectra of MC in the (a) $\omega(782)$, (b) $\phi(1020)$, and (c) ω'/ω'' mass regions. The "Truth-Matched" distribution (green dashed line) corresponds to reconstructed events where the momentum directions of the π^0 and γ_{ISR} successfully match their truth-level counterparts. A loose requirement of $\chi^2_{\text{KF}} < 100$ is applied to the kinematic fit. The continuous spectrum of $\sqrt{s'}$ for the $\pi^+\pi^-\pi^0$ system demonstrates the radiative return mechanism via ISR.

5 Background Study

MC studies show that the main backgrounds include $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$, $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$, and various QED processes.

To verify that these processes constitute the primary backgrounds, a rough estimate is obtained by multiplying the integrated luminosity $\mathcal{L}$ by the hadronic cross section σ_{hadron}. The luminosity values for each data-taking period are listed in Table 1. The background yield is estimated as a function of the $\pi^+\pi^-\pi^0$ invariant mass (m) using the effective luminosity spectrum, which is related to the mass spectrum via:

$$\frac{dL_{\text{eff}}}{dm} = \frac{2m}{s} \cdot F(s, m) \cdot \mathcal{L}_{\text{int}}, \tag{1}$$

where $s = 4E^2$, E is the beam energy, $\mathcal{L}_{\text{int}}$ is the total integrated luminosity, and $F(s, m)$ is the probability function for initial-state photon emission. This function can be written as [22]:

$$\beta = \frac{2\alpha}{\pi}\left[\ln\left(\frac{s}{m_e^2}\right) - 1\right], \tag{2}$$

$$x = 1 - \frac{m^2}{s}, \tag{3}$$

$$\begin{aligned}\frac{dL_{\text{eff}}}{dm} = \frac{2m}{s}\cdot\Bigg\{&\beta x^{\beta-1}\left[1 + \frac{\alpha}{\pi}\left(\frac{\pi^2}{3} - \frac{1}{2}\right) + \frac{3}{4}\beta\right.\\ &\left.+\beta^2\left(\frac{37}{96} - \frac{\pi^2}{12} - \frac{1}{72}\ln\frac{s}{m_e^2}\right)\right]\\ &-\beta\left(1 - \frac{x}{2}\right)\ln\frac{1}{x} - \frac{1 + 3(1-x)^2}{x}\ln(1-x) - 6 + x\Bigg\}\cdot\mathcal{L}_{\text{int}},\end{aligned} \tag{4}$$

where α is the fine-structure constant, $m_e = 0.5110034 \times 10^{-3}$ GeV$/c^2$ is the electron mass, and $\sqrt{s} =$ 3.773 GeV is the center-of-mass energy of the primary e^+e^- system.

The effective luminosity spectrum calculated using this formula for round16 is shown in Fig. 2. Similar calculations were performed for all data-taking rounds.

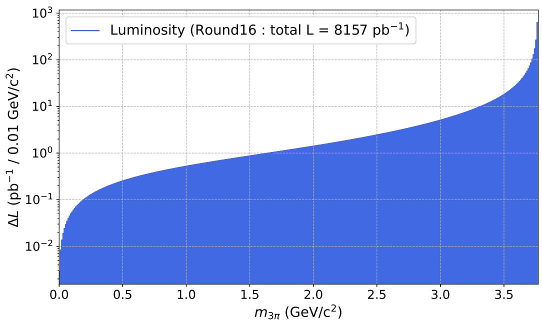


Figure 2: Effective luminosity spectrum as a function of the $\pi^+\pi^-\pi^0$ invariant mass ($m_{3\pi}$) calculated for round16 with total integrated luminosity $\mathcal{L}_{\text{int}} = 8157$ pb^{-1}. The spectrum is shown with logarithmic scale on the vertical axis.

Based on this method, rough yield estimates for the two processes with additional π^0, $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ (non-ISR) and $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ (ISR), were calculated for the four data-taking periods (round03/04, round15, round16, round17). The results are summarized in Table 3. For the ISR channel, the estimation relies on the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ cross section measured by the BaBar collaboration [23].

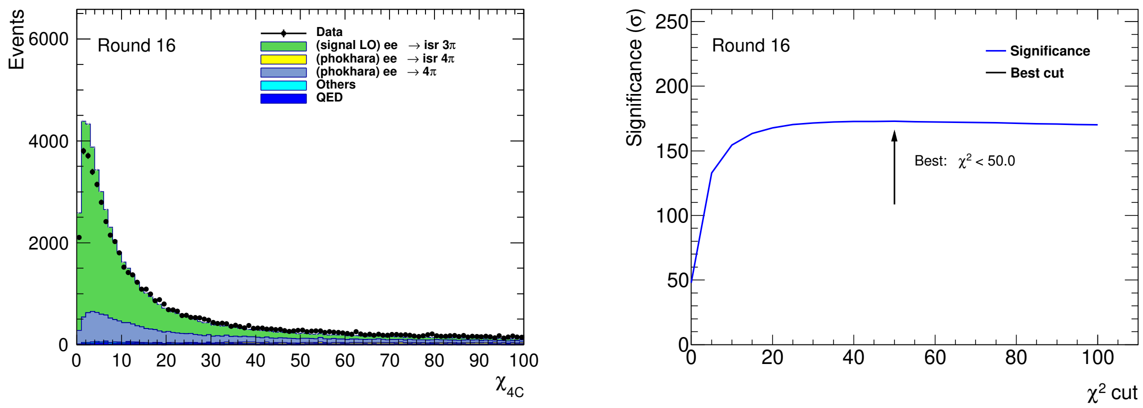
It is important to note that the values listed in the table are initial yield estimates without any selection efficiency corrections; the actual background contribution to the signal region is obtained by multiplying these yields by the corresponding selection efficiency relevant to the specific analysis cuts. These estimates serve only to confirm that these processes constitute the main sources of background and to provide guidance for optimizing subsequent selection criteria.

The final background yield is then obtained by multiplying the effective luminosity spectrum by the cross section and a selection-efficiency factor that varies dynamically with the analysis cuts. This method is used solely to confirm that $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ and $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\text{ISR}}$ are the dominant backgrounds and to optimize the selection criteria. It is not used to derive the final background estimates; precise yields for these two components will be determined using data-driven methods. Based

Table 3: Rough estimates of the expected yields for the two processes with additional π^0 before applying any selection efficiency. The integrated luminosities are taken from Table 1. The cross section for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ is taken as $\sigma = 0.22$ nb at $\sqrt{s} = 3.773$ GeV, while the energy-dependent cross section for the ISR process is derived from the measured $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ cross section [23]. The final background contribution is obtained by multiplying these yields by the corresponding selection efficiency.

Data sample (Run period)	$N(e^+e^- \to \pi^+\pi^-\pi^0\pi^0)$	$N(e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\mathrm{ISR}})$
round03-04	511940	1142096
round15	872209	1945825
round16	1424346	3177597
round17	731817	1632623

on the rough background estimate from $\mathcal{L} \times \sigma_{\mathrm{hadron}}$, the signal yield was approximated by subtracting the background from the data, allowing for a preliminary significance calculation. Using this approach, a figure of merit (FOM) defined as $S/\sqrt{S+B}$ was constructed to evaluate the signal significance. To optimize the event selection, a scan over the kinematic fit χ^2 was performed, with the result for Round 16 shown in Fig. 3. The Round 16 optimization yields an optimal cut of $\chi^2_{\mathrm{KF}} < 50$. As shown in Fig. 4, consistent convergence of significance is also observed for Rounds 03/04, 15, and 17 with the same cut.

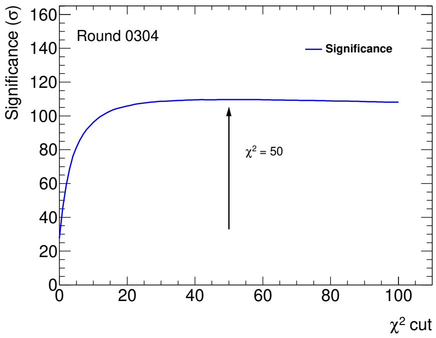


(a) χ^2 distribution for the 4C kinematic fit

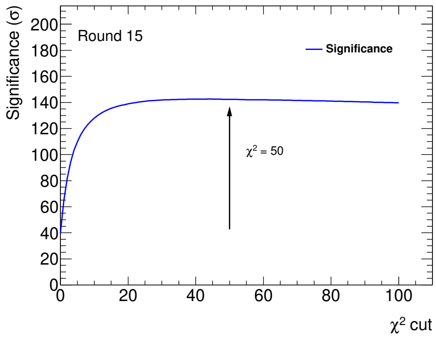
(b) FOM scan as a function of the χ^2 cut

Figure 3: (a) Distribution of the kinematic fit χ^2 for the four-constraint (4C) fit using the Round 16 data sample. (b) The corresponding figure of merit ($S/\sqrt{S+B}$) as a function of the applied χ^2 cut value for the Round 16 data.

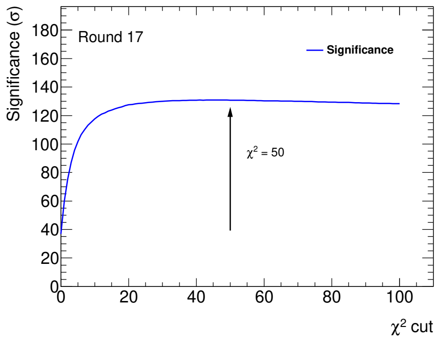
After applying the optimal χ^2_{KF} selection, the invariant mass spectrum of the 3π system ($M_{3\pi}$) can be divided into three distinct regions for further study: the $\omega(782)$ region, the $\phi(1020)$ region, and a higher-mass region associated with possible ω'/ω'' excitations. In the high-mass ω'/ω'' region, a significant background component from events containing additional π^0 mesons was observed. To quantify this, the number of π^0 candidates per event was counted, where any π^0 candidate satisfying a one-constraint



(a) Round 03/04



(b) Round 15



(c) Round 17

Figure 4: FOM ($S/\sqrt{S+B}$) as a function of the applied χ^2 cut value for (a) Round 03/04, (b) Round 15, and (c) Round 17 data samples.

kinematic fit $\chi^2 < 200$ was accepted. The distribution of this π^0 multiplicity is shown in Fig. 5. To suppress this specific background, a requirement of exactly one π^0 candidate per event was applied exclusively in the ω'/ω'' mass region. No such restriction was applied to the $\omega(782)$ and $\phi(1020)$ signal regions. The following studies and distributions are based on the Round 16 data sample.

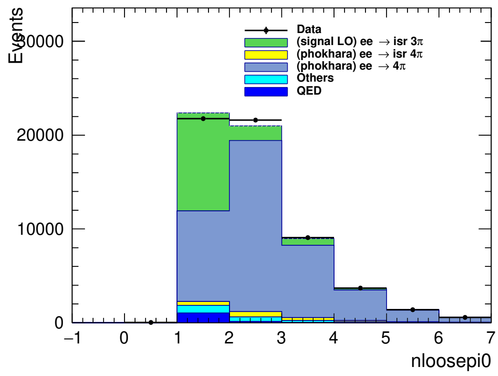
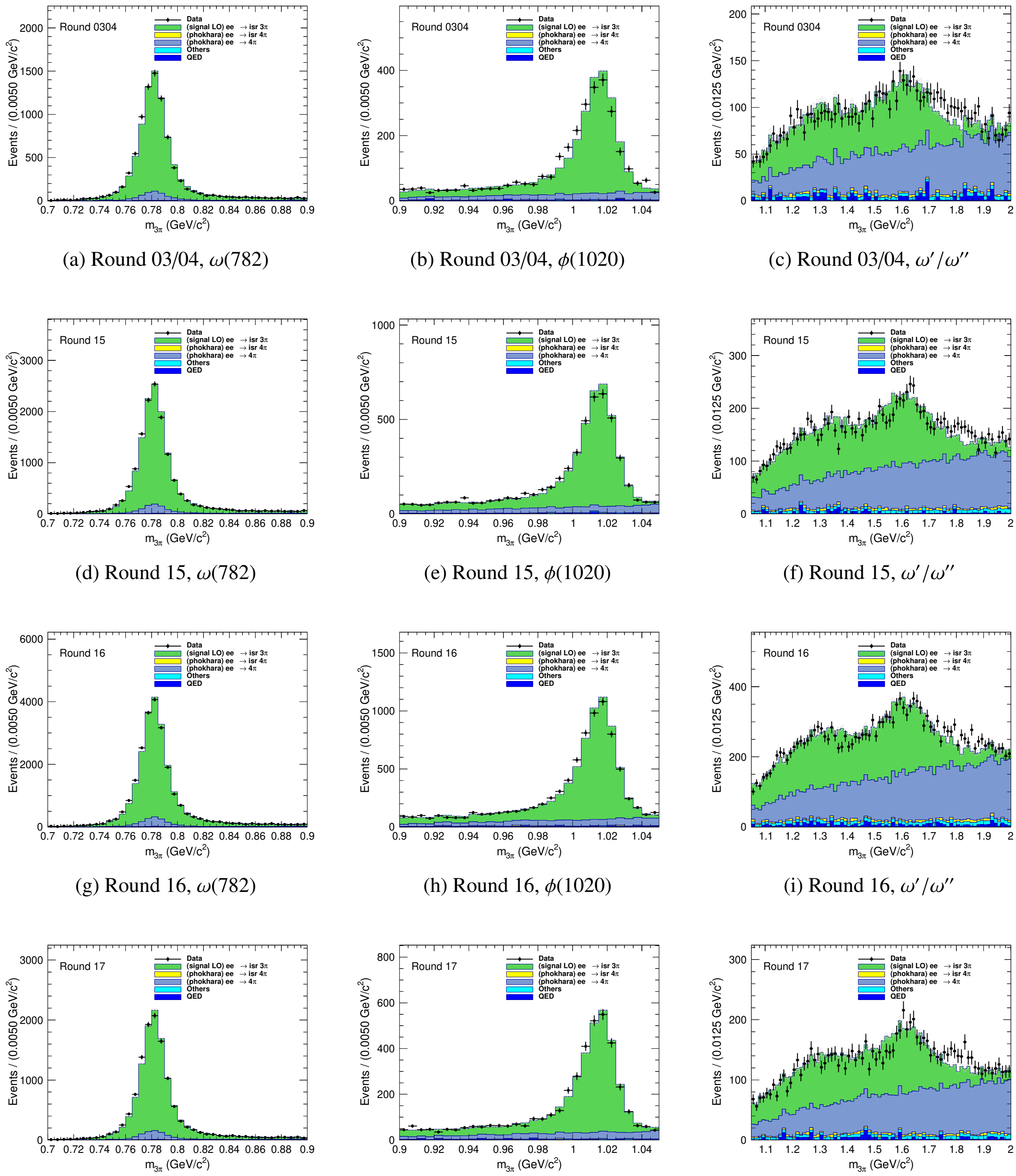


Figure 5: Distribution of the number of π^0 candidates per event in the ω'/ω'' mass region using the Round 16 data. A π^0 candidate is counted if its 1C kinematic fit $\chi^2 < 200$.

The resulting $M_{3\pi}$ invariant mass spectra, after applying all selection criteria including the region-specific π^0 multiplicity cut, are presented separately for the three mass regions in Fig. 6. The $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\mathrm{ISR}}$ contribution is negligible and will not be treated as a separate background in the subsequent fitting procedure; only the non-ISR $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ process will be explicitly estimated using data-driven methods and constrained in the fit.



(a) Round 03/04, $\omega(782)$ (b) Round 03/04, $\phi(1020)$ (c) Round 03/04, ω'/ω''

(d) Round 15, $\omega(782)$ (e) Round 15, $\phi(1020)$ (f) Round 15, ω'/ω''

(g) Round 16, $\omega(782)$ (h) Round 16, $\phi(1020)$ (i) Round 16, ω'/ω''

(j) Round 17, $\omega(782)$ (k) Round 17, $\phi(1020)$ (l) Round 17, ω'/ω''

Figure 6: Invariant mass spectra $M_{3\pi}$ for different data rounds and mass regions, $N_{\pi^0} = 1$ is applied for the ω'/ω'' mass region.

5.1 Estimate of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ backgrounds

It is assumed that the total yield of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events produced in the data is a fixed quantity. This process can contribute background to the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\mathrm{ISR}}$ process when a photon from one of the π^0 decays is misidentified as the initial-state radiation photon (γ_{ISR}). The combined selection proceeds as follows.

First, the full $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ selection is applied:

- **$\pi^+\pi^-\pi^0\pi^0$ Candidate Selection:**
  - Exactly two good charged tracks.
  - nPi0 ≥ 2 and nGoodShowers ≥ 4 (ensuring at least two π^0s).
  - Application of a total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
  - Looping over all combinations of π^0 candidates, the combination with the minimum $\chi^2_{4\mathrm{C}}$ is chosen.
  - A requirement of $\chi^2_{\mathrm{KF}} < 10$ on the kinematic fit.

Subsequently, within the same event, a loose selection for $e^+e^- \to \gamma_{\mathrm{ISR}}\pi^+\pi^-\pi^0$ is applied to reconstruct the 3π invariant mass spectrum:

- **Loose $\gamma_{\mathrm{ISR}}\pi^+\pi^-\pi^0$ Candidate Selection:**
  - The same two good charged tracks.
  - nPi0 ≥ 1 and nGoodShowers ≥ 3 (ensuring at least one π^0 and one photon assigned as γ_{ISR}).
  - Application of the same total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
  - Looping over all combinations of π^0 candidates and EMC showers, the combination with the minimum $\chi^2_{4\mathrm{C}}$ is chosen.
  - No additional kinematic fit χ^2 requirement is applied at this stage.

To estimate the background from 4π events to the 3π signal region, two efficiencies are defined using MC based on the selections above:

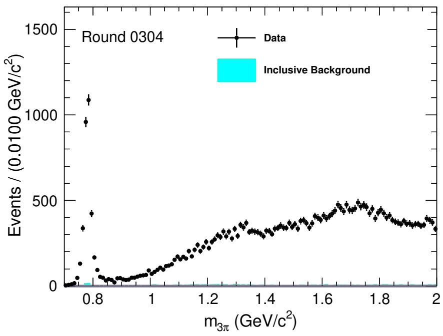
- ϵ_{sig}: the efficiency for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events to pass the nominal 4π selection (thereby allowing the yield of this process in data to be measured), and then to additionally satisfy the loose $\gamma_{\mathrm{ISR}}\pi^+\pi^-\pi^0$ selection for reconstructing the 3π invariant mass spectrum.
- ϵ_{bkg}: the efficiency for the same 4π events to pass the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\mathrm{ISR}}$ selection described in Sec. 4, i.e., to be misreconstructed as background in the 3π mass region.

The observed distribution of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ candidates in data (selected under the nominal 4π criteria) can be converted into an estimate of its background contribution in the signal region using the efficiency ratio $R = \epsilon_{\mathrm{bkg}}/\epsilon_{\mathrm{sig}}$:

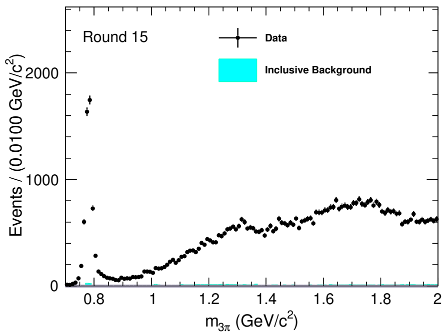
$$N_{\mathrm{bkg}} = R \times N^{\mathrm{obs}}_{4\pi} = \frac{\epsilon_{\mathrm{bkg}}}{\epsilon_{\mathrm{sig}}} \times N^{\mathrm{obs}}_{4\pi}, \tag{5}$$

where $N^{\text{obs}}_{4\pi}$ is the number of observed $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events selected under the nominal 4π criteria, and N_{bkg} is the estimated number of background events from this process in the $\pi^+\pi^-\pi^0$ invariant mass region.

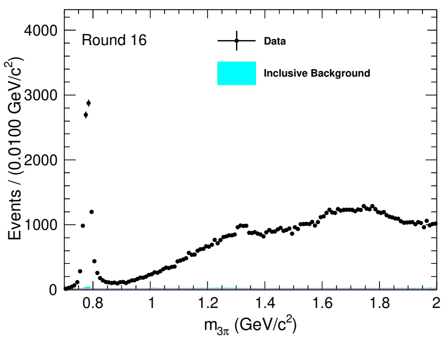
The selection described above results in a highly pure sample of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events. The invariant mass spectrum of the corresponding three-pion system from this pure sample, which serves as the input template for background estimation, is shown in Fig. 8.



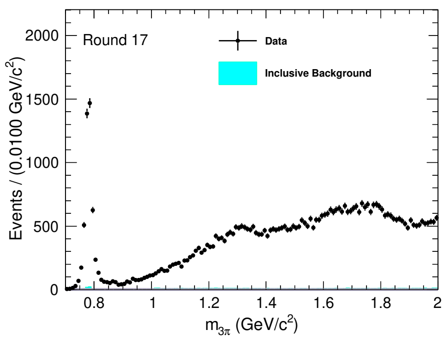
(a) Round 03/04



(b) Round 15



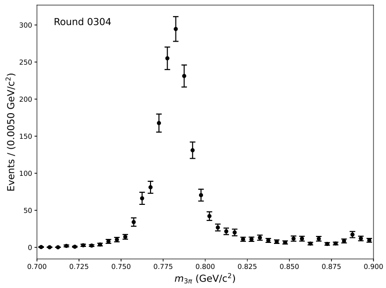
(c) Round 16



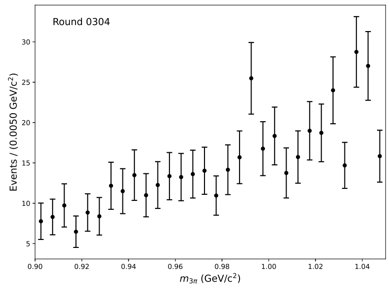
(d) Round 17

Figure 7: The 3π invariant mass spectrum from selected $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events in the (a) Round 03/04, (b) Round 15, (c) Round 16, and (d) Round 17 datasets. These distributions, obtained from data after applying the dual-tag selection criteria for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$, represent the pure signal samples used as templates for background estimation.

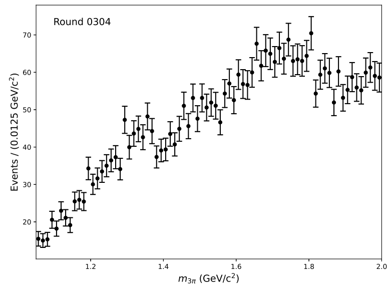
The observed distribution in the data, after applying the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ selection criteria, will be scaled by the efficiency ratio $R = \epsilon_{\text{bkg}}/\epsilon_{\text{sig}}$ to estimate the final background contribution of this process in the signal region of the measurement channel. As established and shown in Fig. 7, the background contamination within this selected sample is negligible. The resulting background-subtracted signal distributions in the three 3π invariant mass regions are presented in Fig. 8. All other backgrounds will be subtracted directly.



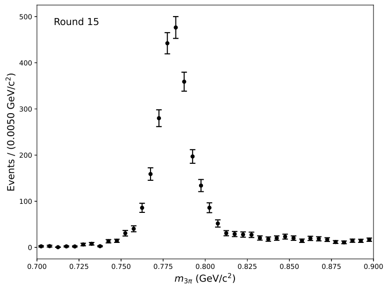
(a) Round 03/04, $\omega(782)$ region



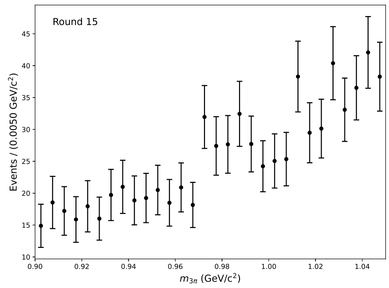
(b) Round 03/04, $\phi(1020)$ region



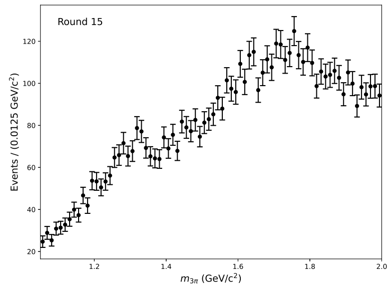
(c) Round 03/04, ω'/ω'' region



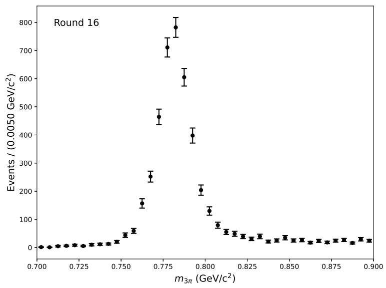
(d) Round 15, $\omega(782)$ region



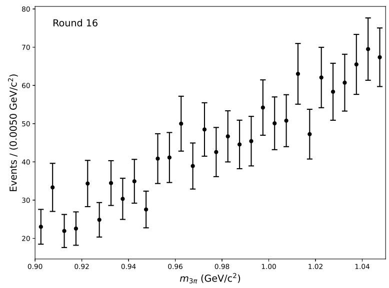
(e) Round 15, $\phi(1020)$ region



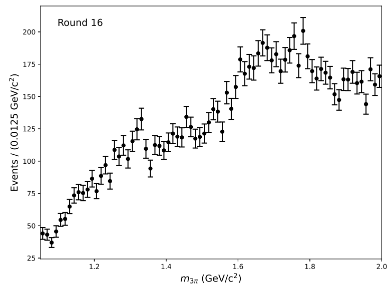
(f) Round 15, ω'/ω'' region



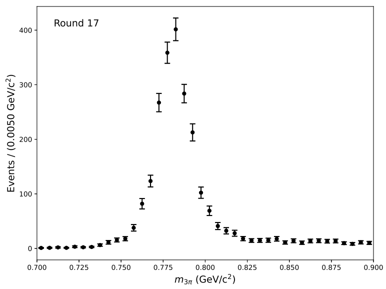
(g) Round 16, $\omega(782)$ region



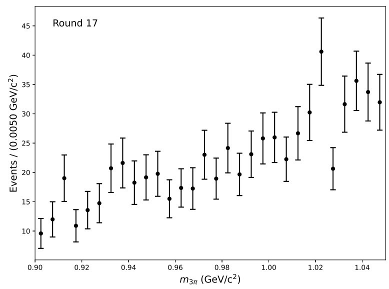
(h) Round 16, $\phi(1020)$ region



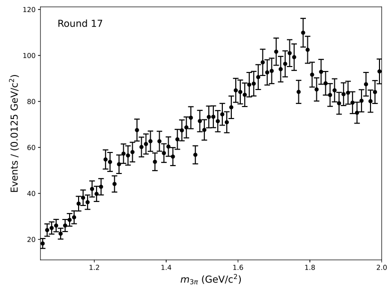
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 8: Estimated background contributions from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ in the signal region after scaling, shown for four data-taking periods. Each row corresponds to a different round: (a)–(c) Round 03/04, (d)–(f) Round 15, (g)–(i) Round 16, and (j)–(l) Round 17. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right).

5.2 Estimate of Other Backgrounds and Signal Yields

The estimation of the dominant peaking backgrounds from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ has been completed in the previous subsection. These backgrounds exhibit shapes similar to the signal resonance peaks and are therefore categorized as peaking backgrounds.

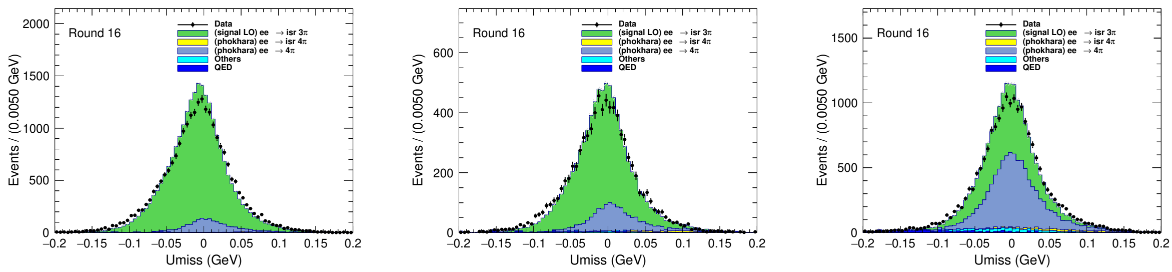
In addition to the peaking component, a non-peaking continuum background component must also be accounted for in the analysis. The primary objective of this study is to measure the cross section for $e^+e^- \to \pi^+\pi^-\pi^0$ via ISR. The ISR photon reduces the effective c.m. energy, allowing the measurement of the signal yield in narrow intervals of the 3π invariant mass ($M_{3\pi}$).

The $M_{3\pi}$ spectrum is divided into three physically motivated regions, each subdivided into multiple bins for detailed study:

- The $\omega(782)$ region (0.7–0.9 GeV$/c^2$), divided into 40 bins.
- The $\phi(1020)$ region (0.9–1.05 GeV$/c^2$), divided into 30 bins.
- The ω'/ω'' region (1.05–2.0 GeV$/c^2$), divided into 76 bins.

For each small $M_{3\pi}$ interval, the signal yield is extracted through an extended unbinned maximum-likelihood fit to the $U_{\rm miss}$ distribution, where $U_{\rm miss} = E_{\rm miss} - |\vec{p}_{\rm miss}|$ is computed from the missing four-momentum after removing the π^+, π^-, π^0, and $\gamma_{\rm ISR}$ from the initial e^+e^- system. These quantities are calculated using the measured momenta without any kinematic fit update. The fit model incorporates probability density functions (PDFs) representing the signal and all background components.

To illustrate the composition of $U_{\rm miss}$ distributions across different $M_{3\pi}$ regions, the inclusive MC simulation is examined. Figure 9 shows the $U_{\rm miss}$ distributions for the full $\omega(782)$ region, the full $\phi(1020)$ region, and the full ω'/ω'' region respectively, with the peaking background contribution from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ stacked according to its expected yield. These distributions demonstrate the characteristic features of each mass region that motivate our fitting strategy.



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω' and ω'' region

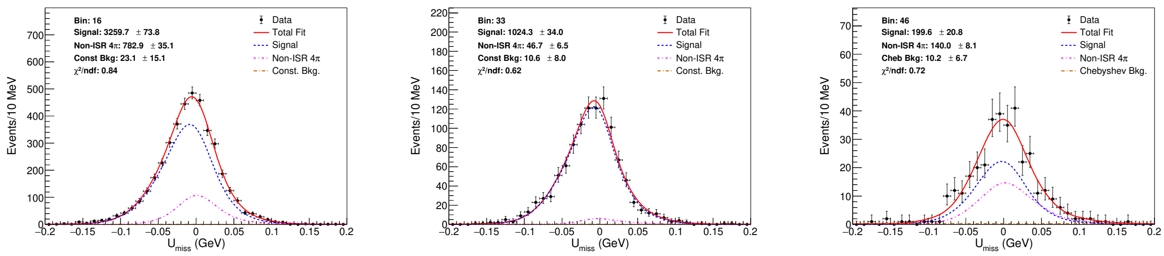
Figure 9: $U_{\rm miss}$ distributions from inclusive MC simulation for three $M_{3\pi}$ regions: (a) the $\omega(782)$ region, (b) the $\phi(1020)$ region, and (c) the higher-mass ω'/ω'' region.

Based on the understanding from the inclusive MC, the PDFs for the fit are constructed as follows:

- **Signal PDF:** Histogram shape from signal MC, convolved with a Gaussian for resolution correction. The $\chi^2_{\rm KF}$ used in the $\chi^2_{\rm KF} < 50$ selection criterion has been corrected for data-MC differences in helix parameters as described in Section 6.5.

- **Peaking background PDF:** Shape for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ from dedicated MC, convolved with the same Gaussian as signal. Yield constrained by Gaussian term using data-driven estimate from Section 5.
- **PDF shape sharing across bins:** Limited MC statistics preclude bin-by-bin shape extraction. Instead:
  - For the $\omega(782)$ and $\phi(1020)$ regions, where the $M_{3\pi}$ bin width is 5 MeV/c^2, the PDF shapes are extracted from MC simulation every five consecutive bins, corresponding to 25 MeV/c^2 intervals. All five bins within each such interval share the same PDF shape, under the assumption that the U_{miss} distribution varies slowly over this mass range.
  - For the ω'/ω'' region, where the $M_{3\pi}$ bin width is 12.5 MeV/c^2, the PDF shapes are also extracted every 25 MeV/c^2. Each extracted shape is therefore shared by two adjacent bins, balancing the need for shape precision against the limited MC statistics in this higher-mass region.
- **Continuum background PDF:** All remaining minor background processes are collectively described by a polynomial function. A first-order polynomial (linear function) is used for the $\omega(782)$ and $\phi(1020)$ regions, while a second-order Chebyshev polynomial is employed for the ω'/ω'' region, as suggested by the shape of the inclusive MC distribution in Fig. 9c.

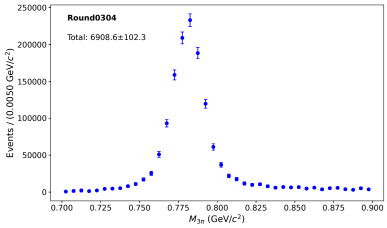
The fit procedure is validated by examining the quality of the fits to data in representative $M_{3\pi}$ bins across the three mass regions, using Round 16 as an illustrative example. Figure 10 presents typical fit results for three selected bins, one from each region, demonstrating the ability of the fit model to accurately describe the data. The signal component, peaking background from $\pi^+\pi^-\pi^0\pi^0$, and polynomial continuum background are clearly distinguished, and the total PDF provides an excellent description of the data distribution in each case.



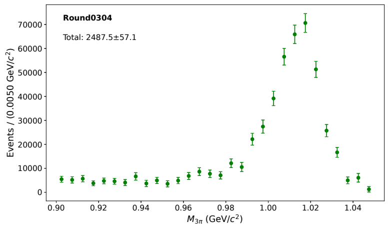
(a) Example bin in the $\omega(782)$ region (b) Example bin in the $\phi(1020)$ region (c) Example bin in the ω'/ω'' region

Figure 10: Typical U_{miss} fit results for representative $M_{3\pi}$ bins from each of the three mass regions, shown for Round 16 data.

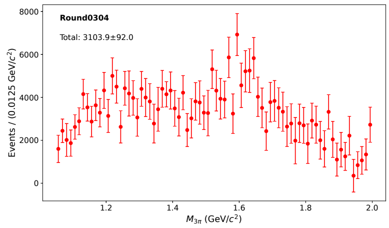
Applying this fitting procedure to all $M_{3\pi}$ bins yields the extracted signal yields across the entire mass spectrum. The same fit is performed independently for four data-taking periods: Round 03/04, Round 15, Round 16, and Round 17. Figure 11 presents the measured signal yields as a function of $M_{3\pi}$ for all four periods, organized by the three mass regions. In each panel, the $\omega(782)$ resonance is clearly visible as a prominent peak around 0.78 GeV$/c^2$, while the $\phi(1020)$ resonance appears as a distinct but smaller peak near 1.02 GeV$/c^2$. In the higher mass regions, broad structures corresponding to the ω' and ω'' excitations are observed, with significantly reduced yields due to the decreasing cross section and lower ISR photon probability at larger momentum transfers. The results show good consistency across the four data-taking periods, with statistical uncertainties represented by the error bars.



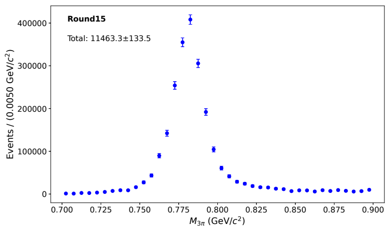
(a) Round 03/04, $\omega(782)$ region



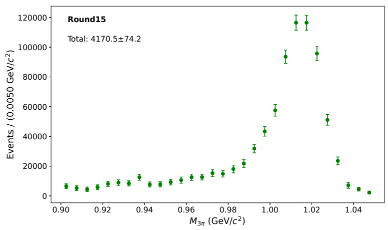
(b) Round 03/04, $\phi(1020)$ region



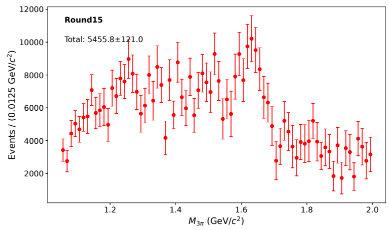
(c) Round 03/04, ω'/ω'' region



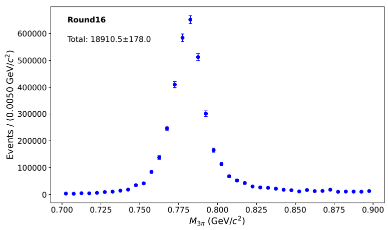
(d) Round 15, $\omega(782)$ region



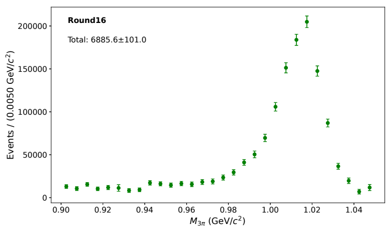
(e) Round 15, $\phi(1020)$ region



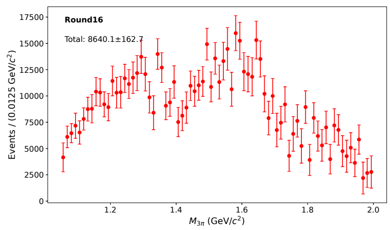
(f) Round 15, ω'/ω'' region



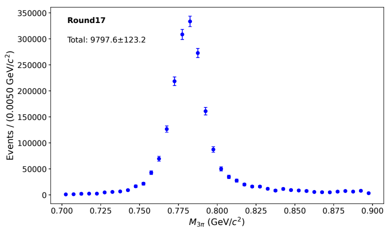
(g) Round 16, $\omega(782)$ region



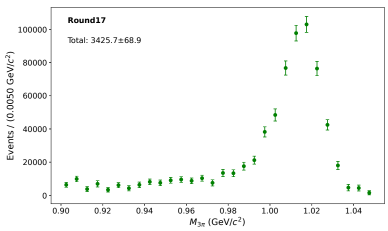
(h) Round 16, $\phi(1020)$ region



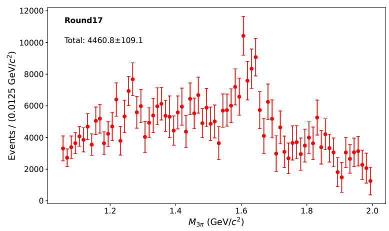
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 11: Extracted signal yields as a function of $M_{3\pi}$ for the three mass regions, shown separately for four data-taking periods. Each row corresponds to a different round: (a)–(c) Round 03/04, (d)–(f) Round 15, (g)–(i) Round 16, and (j)–(l) Round 17. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right). Error bars represent the statistical uncertainties from the fits.

6 Correction of Signal Efficiency

To extract the dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$, we need the generator-level yield $\left(\frac{dN}{dm}\right)^{\text{gen}}$, which is related to the cross section within each mass interval by:

$$\left(\frac{dN}{dm}\right)^{\text{gen}} = \sigma_{3\pi}^{\text{dress}}(m) \cdot \frac{dL}{dm} \cdot \varepsilon \cdot \mathcal{B}(\pi^0 \to \gamma\gamma), \tag{6}$$

where $\frac{dL}{dm}$ is the integrated luminosity in that interval, and $\mathcal{B}(\pi^0 \to \gamma\gamma)$ is the branching fraction of the $\pi^0 \to \gamma\gamma$ decay.

From data, we have obtained the observed signal yield $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$ in each mass interval by fitting the U_{miss} distribution, as shown in Fig. 11. Due to detector effects, events can migrate between different mass intervals, so these observed yields do not directly give the generator-level yields in Eq. (6). The migration is described by the response matrix P_{ij}, which relates the observed yield in bin i to the generator-level yield in bin j:

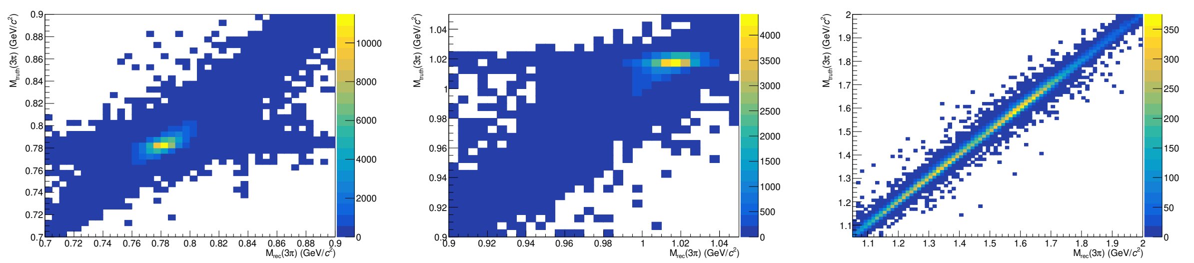
$$\left(\frac{dN}{dm}\right)_i^{\text{obs}} = \sum_j P_{ij} \left(\frac{dN}{dm}\right)_j^{\text{gen}}, \tag{7}$$

The generator-level yields $\left(\frac{dN}{dm}\right)_j^{\text{gen}}$ are therefore obtained by solving Eq. (7) for $\left(\frac{dN}{dm}\right)_j^{\text{gen}}$ given the observed yields $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$ and the response matrix P_{ij}. This numerical procedure is referred to as unfolding.

Let A_{ij} denote the number of MC events that are generated in true bin j and reconstructed in bin i, where i and j index the observed and generator-level mass bins, respectively. The migration matrices A_{ij} for three representative mass regions are shown in Fig. 12, illustrating how events generated in a given true bin are distributed across observed bins due to detector effects. This matrix is then normalized column-wise to obtain the response matrix used in Eq. (7):

$$P_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_d} A_{kj}}, \qquad \sum_{i=1}^{n_d} P_{ij} = 1, \tag{8}$$

where n_d is the number of reconstructed bins.



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω'/ω'' region

Figure 12: Response matrices A_{ij} for three representative mass regions, showing the migration of events between generator-level and reconstructed $M_{3\pi}$ bins. The diagonal structure indicates good resolution, while off-diagonal entries reflect the effects of finite detector resolution and acceptance.

Since the unfolding procedure relies on the response matrix derived from MC simulation, any discrepancies between data and MC can lead to biased results. Therefore, the MC sample used to construct the response matrix must be corrected to ensure that it accurately reflects the data. These discrepancies arise from various sources: the tracking efficiency of charged particles, PID performance, the reconstruction efficiency of neutral π^0 mesons, the detection efficiency of ISR photons, differences in the production rate of additional ISR photons (NLO) between the Phokhara generator and data, and the discrepancy in the χ^2_{4C} distribution between data and MC. The following subsections address each of these corrections in detail.

6.1 Tracking and PID efficiencies

The tracking and PID efficiencies for charged pions are adopted from the dedicated study in Ref. [25], where they are measured using $D^0\bar{D}^0$ and D^+D^- events. The kinematic coverage of that study matches the momentum range of charged pions in this analysis. For each momentum bin, the correction factors $c_{\mathrm{trk}}(p)$ and $c_{\mathrm{pid}}(p)$ are derived separately for tracking and PID.

6.2 The π^0 reconstruction efficiency

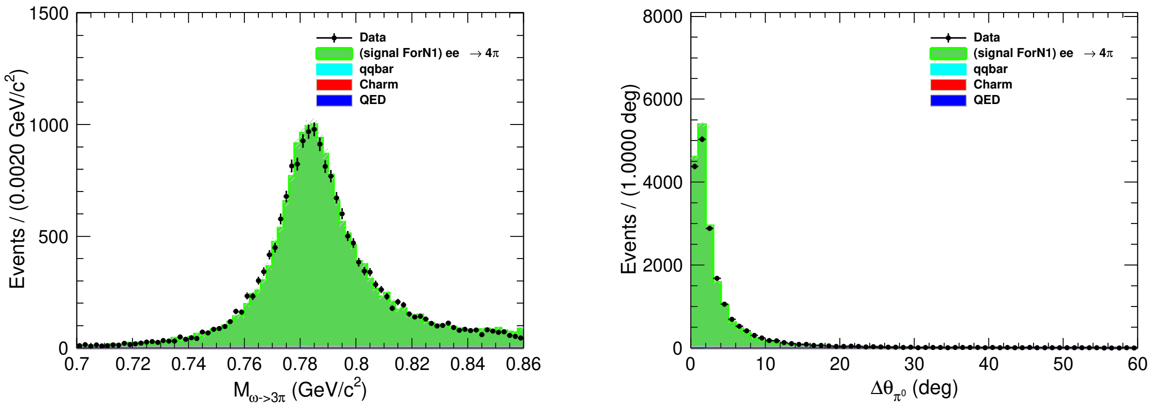
The reconstruction efficiency for π^0 mesons is studied using a dedicated control sample of $e^+e^- \to \omega\pi^0_{\mathrm{tag}}$ events at $\sqrt{s} = 3.773\,\mathrm{GeV}$, with $\omega \to \pi^+\pi^-\pi^0_{\mathrm{pred}}$. This process provides a clean environment to study π^0 efficiency due to the distinct kinematic properties of the two neutral pions: the π^0_{tag} carries approximately half of the center-of-mass momentum, while the π^0_{pred} from ω decay has a momentum spectrum ranging from 0 to $\sqrt{s}/2$, making it well suited for a momentum-dependent efficiency study.

The measurement is performed independently for each data-taking period (rounds 0304, 15, 16, and 17) using the 3.773 GeV data sample. The selection criteria are as follows:

- The selection requirements for charged pions and π^0 reconstruction are identical to those used in the signal channel $e^+e^- \to \pi^+\pi^-\pi^0$.
- Events are required to have at least one reconstructed π^0 candidate.
- In events with multiple π^0 candidates, the two with the best fit quality from 1C kinematic fitting are retained for further analysis.
- The π^0_{tag} is identified as the π^0 candidate whose energy is closest to half the center-of-mass energy ($\sqrt{s}/2$), with a requirement that the difference be within 0.3 GeV.
- A 1C kinematic fit is then performed: the recoil momentum of the $\pi^+\pi^-\pi^0_{\mathrm{tag}}$ system against the initial e^+e^- system is calculated in the center-of-mass frame; this recoil momentum is constrained to the π^0 mass, corresponding to the missing π^0_{pred}; subsequently, the $\pi^+\pi^-$ system together with the π^0_{tag} is constrained to the center-of-mass system.

The 1C kinematic fit is required to satisfy $\chi^2 < 15$. After the fit, the four-momentum of the π^0_{pred} is updated. The invariant mass of the $\pi^+\pi^-\pi^0_{\mathrm{pred}}$ system is then calculated, and events are required to have this invariant mass within the ω meson mass window of 0.7–0.86 GeV.

Before applying the momentum-dependent efficiency study, we first examine the overall $\pi^+\pi^-\pi^0_{\rm pred}$ invariant mass distribution without requiring a matching reconstructed π^0 in the predicted direction. Figure 13a shows this distribution for all events passing the selection criteria, with data, MC, and estimated background contributions overlaid. The signal region 0.7–0.86 GeV exhibits a clean peak with negligible background, confirming the purity of the control sample.



(a) (b)

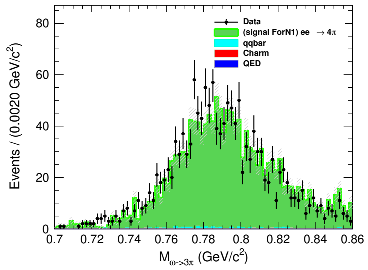
Figure 13: (a) Full $\pi^+\pi^-\pi^0_{\rm pred}$ invariant mass distribution for all selected events, showing data (points), MC (green), and estimated background (other colors). The vertical dashed lines indicate the ω mass window used for event selection. (b) Angular separation $\Delta\theta$ between the predicted $\pi^0_{\rm pred}$ direction and additional reconstructed π^0 candidates. The vertical dashed line indicates the 20° matching cone used in this analysis.

To associate a reconstructed π^0 with the predicted $\pi^0_{\rm pred}$, we examine the angular separation $\Delta\theta$ between the direction of the predicted $\pi^0_{\rm pred}$ and any additional reconstructed π^0 candidate (excluding the $\pi^0_{\rm tag}$). Figure 13b shows the distribution of this angular separation for events where an additional π^0 is found. A clear peak at small angles is observed, corresponding to correctly matched π^0 candidates. Based on this distribution, we define a matching cone of 20°: if an additional reconstructed π^0 lies within 20° of the predicted direction, it is considered as the reconstructed $\pi^0_{\rm pred}$.

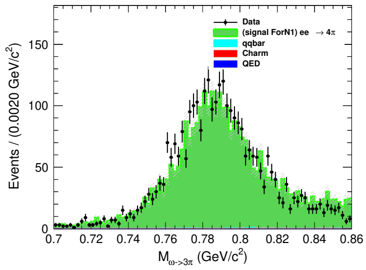
For each momentum bin, Fig. 14 shows the $\pi^+\pi^-\pi^0_{\rm pred}$ invariant mass distribution for events where a matching reconstructed π^0 is found within the 20° cone. The distributions for $p_{\pi^0} < 0.2$ GeV, $0.2 \le p_{\pi^0} < 0.4$ GeV, $0.4 \le p_{\pi^0} < 0.6$ GeV, $0.6 \le p_{\pi^0} < 0.8$ GeV, $0.8 \le p_{\pi^0} < 1.0$ GeV, $1.0 \le p_{\pi^0} < 1.2$ GeV, $1.2 \le p_{\pi^0} < 1.4$ GeV, and $p_{\pi^0} \ge 1.4$ GeV all show clean signal peaks, demonstrating that the matching procedure is effective for the entire momentum range.

Given the negligible background in the signal region, the π^0 reconstruction efficiency can be directly determined by counting events. For each momentum bin, we define:

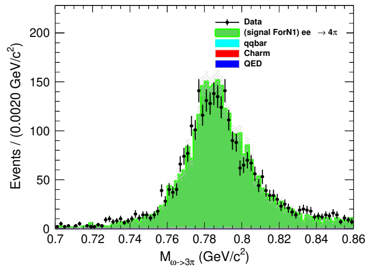
- $N^{\rm total}_{\rm data}(p)$: number of data events in the ω mass window, regardless of whether a matching π^0 is found.
- $N^{\rm matched}_{\rm data}(p)$: number of data events in the ω mass window with a matching reconstructed π^0 within 20° of the predicted direction.



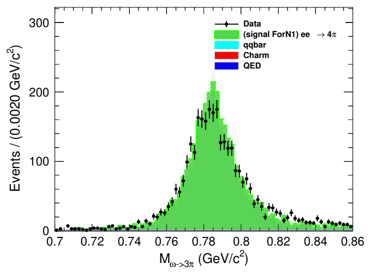
(a) Bin 0: $p < 0.2\,\text{GeV}$



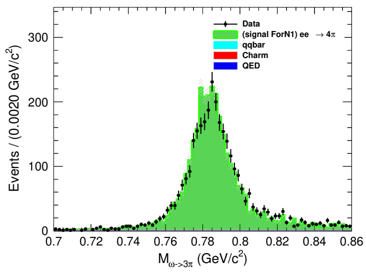
(b) Bin 1: $0.2 \leq p < 0.4\,\text{GeV}$



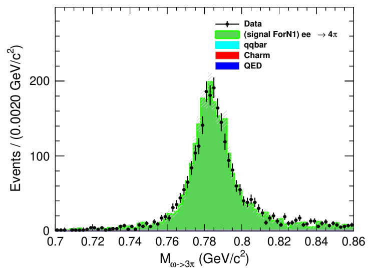
(c) Bin 2: $0.4 \leq p < 0.6\,\text{GeV}$



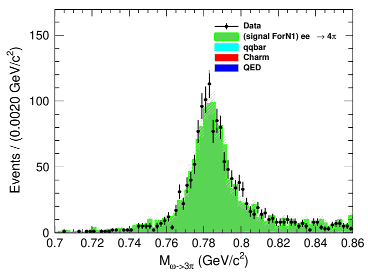
(d) Bin 3: $0.6 \leq p < 0.8\,\text{GeV}$



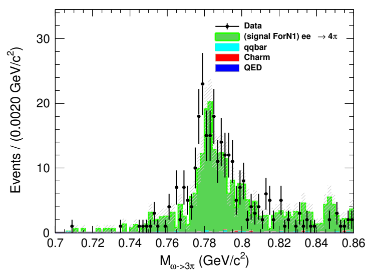
(e) Bin 4: $0.8 \leq p < 1.0\,\text{GeV}$



(f) Bin 5: $1.0 \leq p < 1.2\,\text{GeV}$



(g) Bin 6: $1.2 \leq p < 1.4\,\text{GeV}$



(h) Bin 7: $p \geq 1.4\,\text{GeV}$

Figure 14: $\pi^+\pi^-\pi^0_{\text{pred}}$ invariant mass distributions for events with a matching reconstructed π^0 within 20°, shown for each π^0_{pred} momentum bin. Data (points), MC (green), and estimated background (other colors) are overlaid. The vertical dashed lines indicate the ω mass window used for event selection.

- $N^{\text{total}}_{\text{MC}}(p)$ and $N^{\text{matched}}_{\text{MC}}(p)$: corresponding quantities for MC.

The efficiencies in data and MC are then computed as:

$$\varepsilon_{\text{data}}(p) = \frac{N^{\text{matched}}_{\text{data}}(p)}{N^{\text{total}}_{\text{data}}(p)}, \quad \varepsilon_{\text{MC}}(p) = \frac{N^{\text{matched}}_{\text{MC}}(p)}{N^{\text{total}}_{\text{MC}}(p)}. \tag{9}$$

The statistical uncertainty on each efficiency is given by the binomial error:

$$\sigma_{\varepsilon} = \sqrt{\frac{\varepsilon(1-\varepsilon)}{N_{\text{total}}}}, \tag{10}$$

where N_{total} is the total number of events in the corresponding sample.

The correction factor for π^0 reconstruction is then derived as the ratio of data efficiency to MC efficiency:

$$c_{\pi^0}(p) = \frac{\varepsilon_{\text{data}}(p)}{\varepsilon_{\text{MC}}(p)}. \tag{11}$$

The calculated efficiencies and correction factors are presented in Table 4. Each entry is shown as central value ± its statistical uncertainty.

Table 4: The π^0 reconstruction efficiencies in data and MC, and the derived correction factors with their statistical uncertainties for different π^0 momentum intervals.

p_{π^0} (GeV)	$\varepsilon_{\text{data}}$	ε_{MC}	c_{π^0}
< 0.2	0.3892 ± 0.0080	0.3958 ± 0.0064	0.9833 ± 0.0254
0.2 – 0.4	0.4579 ± 0.0061	0.4782 ± 0.0050	0.9575 ± 0.0168
0.4 – 0.6	0.5442 ± 0.0067	0.5636 ± 0.0059	0.9656 ± 0.0163
0.6 – 0.8	0.6265 ± 0.0068	0.6471 ± 0.0060	0.9682 ± 0.0150
0.8 – 1.0	0.6946 ± 0.0068	0.7124 ± 0.0060	0.9750 ± 0.0142
1.0 – 1.2	0.7540 ± 0.0074	0.7785 ± 0.0064	0.9685 ± 0.0139
1.2 – 1.4	0.8089 ± 0.0094	0.8161 ± 0.0082	0.9912 ± 0.0167
≥ 1.4	0.8017 ± 0.0215	0.8574 ± 0.0155	0.9350 ± 0.0283

6.3 The ISR photon detection efficiency

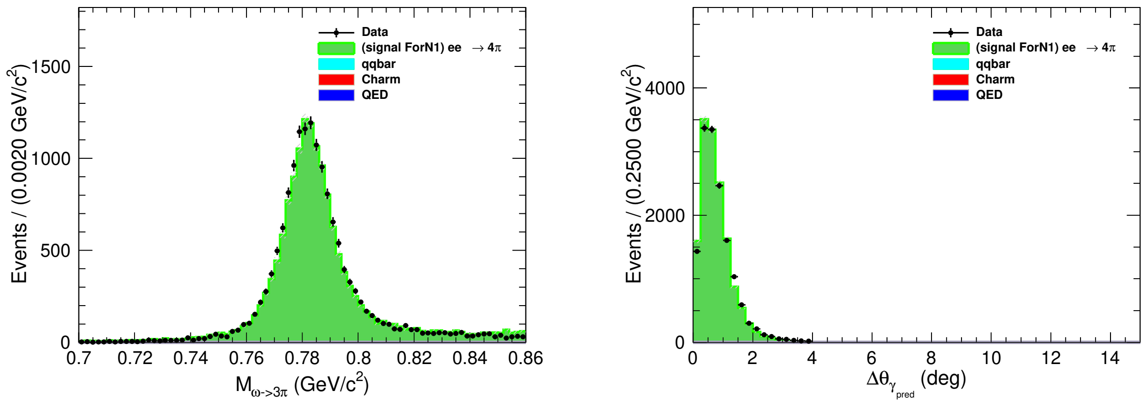
The ISR photon detection efficiency is studied using the same control sample of $e^+e^- \to \omega\pi^0$ events at $\sqrt{s} = 3.773$ GeV, with $\omega \to \pi^+\pi^-\pi^0$. Unlike the previous π^0 efficiency study which tagged one π^0 and studied the other from ω decay, here we identify one π^0 as coming from ω decay and study the photon detection efficiency using the two photons from the decay of the remaining π^0. This remaining π^0, which balances the ω in the event, carries approximately half of the center-of-mass energy ($\sqrt{s}/2$).

The measurement is performed independently for each data-taking period using the 3.773 GeV data sample. The selection criteria are as follows:

- The selection requirements for charged pions are identical to those used in the signal channel $e^+e^- \to \pi^+\pi^-\pi^0$.
- Events are required to have at least one reconstructed π^0 candidate.
- For each reconstructed π^0 candidate, the invariant mass of the $\pi^+\pi^-$ system combined with this π^0 is calculated. Among all π^0 candidates, the one that yields an invariant mass closest to the nominal ω mass and within ± 0.3 GeV is identified as originating from the ω decay (π^0_ω).
- The other π^0 in the event (denoted as $\pi^0_{\rm bal}$) balances the ω system and has a characteristic energy of approximately $\sqrt{s}/2$. It decays into two photons: $\pi^0_{\rm bal} \to \gamma\gamma$. Note that this $\pi^0_{\rm bal}$ itself is not required to be reconstructed; we only study its decay photons, and its existence is inferred from the ω system and momentum conservation.
- All remaining photon candidates in the event (excluding those used in π^0_ω reconstruction) are considered as potential tagged photons. For each such photon candidate, we treat it as the tagged photon ($\gamma_{\rm tag}$) and perform a 2C kinematic fit where the other photon ($\gamma_{\rm pred}$) is inferred via momentum conservation.
- A 2C kinematic fit is performed for each photon candidate with the following constraints:
  - The invariant mass of the missing momentum (corresponding to $\gamma_{\rm pred}$) is constrained to zero (photon mass).
  - The invariant mass of the tagged photon and the predicted photon is constrained to the π^0 mass.
  - The total four-momentum of the event is constrained to the center-of-mass system ($\sqrt{s}$ = 3.773 GeV).
- Entries with $\chi^2 < 5$ from the kinematic fit are classified as good entries. Events are required to have at least one good entry.
- For events with multiple good entries, the two with the smallest χ^2 are retained. From these two, the entry with the smaller tagged photon energy (correspondingly larger predicted photon energy) is selected for further analysis. This selection ensures that the predicted photon energy lies in the region of interest above 1 GeV.
- The selected entry is required to satisfy the kinematic fit quality $\chi^2 < 5$.

Before applying the energy-dependent efficiency study, we first examine the overall $\pi^+\pi^-\pi^0_\omega$ invariant mass distribution without requiring a matching reconstructed photon in the predicted direction. Figure 15a shows this distribution for all events passing the selection criteria, with data, MC, and estimated background contributions overlaid. The signal region 0.7–0.86 GeV exhibits a clean peak with negligible background, confirming the purity of the control sample.

To associate a reconstructed photon with the predicted photon $\gamma_{\rm pred}$, we examine the angular separation $\Delta\theta$ between the direction of the predicted photon and any reconstructed photon candidate (excluding those used in π^0_ω reconstruction). Figure 15b shows the distribution of this angular separation for events



(a) (b)

Figure 15: (a) Full $\pi^+\pi^-\pi^0_\omega$ invariant mass distribution for all selected events, showing data (points), MC (green), and estimated background (other colors). The vertical dashed lines indicate the ω mass window used for event selection. (b) Angular separation $\Delta\theta$ between the predicted photon γ_{pred} direction and reconstructed photon candidates. The vertical dashed line indicates the 4° matching cone used in this analysis.

where a photon is found. A clear peak at small angles is observed, corresponding to correctly matched photon candidates. Based on this distribution, we define a matching cone of $4\circ$: if a reconstructed photon lies within 4° of the predicted direction, it is considered as the reconstructed γ_{pred}.

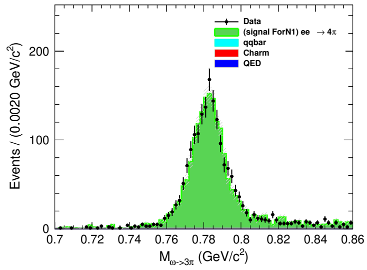
For each energy bin, Fig. 16 shows the $\pi^+\pi^-\pi^0_\omega$ invariant mass distribution for events where a matching reconstructed photon is found within the 4° cone. The distributions for $1.0 \le E_\gamma < 1.1$ GeV, $1.1 \le E_\gamma < 1.2$ GeV, $1.2 \le E_\gamma < 1.3$ GeV, $1.3 \le E_\gamma < 1.4$ GeV, $1.4 \le E_\gamma < 1.5$ GeV, $1.5 \le E_\gamma < 1.6$ GeV, $1.6 \le E_\gamma < 1.7$ GeV, and $E_\gamma \ge 1.8$ GeV all show clean signal peaks, demonstrating that the matching procedure is effective for the entire energy range.

Given the negligible background in the signal region, the photon detection efficiency can be directly determined by counting events. For each energy bin, we define:

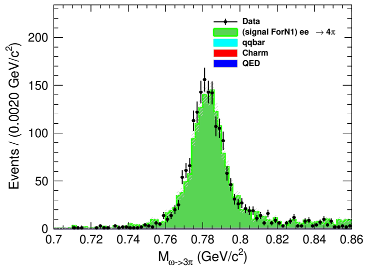
- $N^{\text{total}}_{\text{data}}(E)$: number of data events in the ω mass window, regardless of whether a matching photon is found.
- $N^{\text{matched}}_{\text{data}}(E)$: number of data events in the ω mass window with a matching reconstructed photon within 4° of the predicted direction.
- $N^{\text{total}}_{\text{MC}}(E)$ and $N^{\text{matched}}_{\text{MC}}(E)$: corresponding quantities for MC.

The efficiencies in data and MC are then computed as:

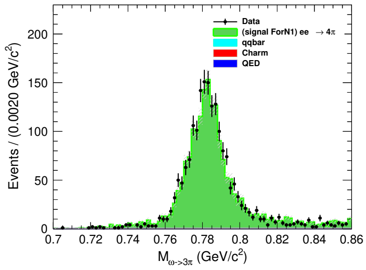
$$\varepsilon_{\text{data}}(E) = \frac{N^{\text{matched}}_{\text{data}}(E)}{N^{\text{total}}_{\text{data}}(E)}, \quad \varepsilon_{\text{MC}}(E) = \frac{N^{\text{matched}}_{\text{MC}}(E)}{N^{\text{total}}_{\text{MC}}(E)}. \tag{12}$$



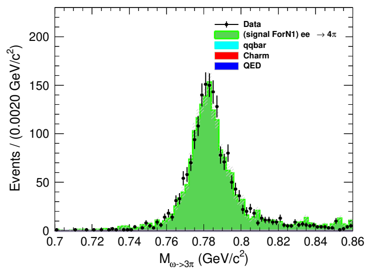
(a) Bin 1: $1.0 \leq E < 1.1\,\mathrm{GeV}$



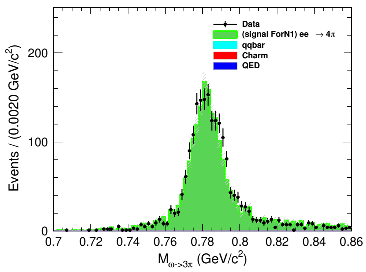
(b) Bin 2: $1.1 \leq E < 1.2\,\mathrm{GeV}$



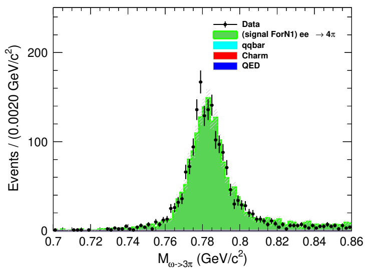
(c) Bin 3: $1.2 \leq E < 1.3\,\mathrm{GeV}$



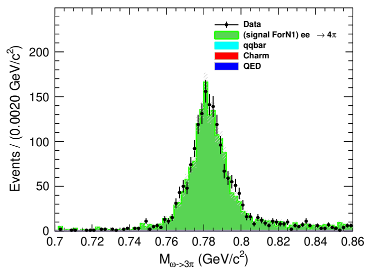
(d) Bin 4: $1.3 \leq E < 1.4\,\mathrm{GeV}$



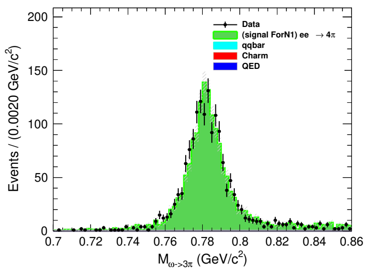
(e) Bin 5: $1.4 \leq E < 1.5\,\mathrm{GeV}$



(f) Bin 6: $1.5 \leq E < 1.6\,\mathrm{GeV}$



(g) Bin 7: $1.6 \leq E < 1.7\,\mathrm{GeV}$



(h) Bin 8: $E \geq 1.8\,\mathrm{GeV}$

Figure 16: $\pi^+\pi^-\pi^0_\omega$ invariant mass distributions for events with a matching reconstructed photon within 4°, shown for each predicted photon energy bin. Data (points), MC (green), and estimated background (other colors) are overlaid. The vertical dashed lines indicate the ω mass window used for event selection.

The statistical uncertainty on each efficiency is given by the binomial error:

$$\sigma_\varepsilon = \sqrt{\frac{\varepsilon(1-\varepsilon)}{N_{\text{total}}}}, \tag{13}$$

where N_{total} is the total number of events in the corresponding sample.

The correction factor for photon detection is then derived as the ratio of data efficiency to MC efficiency:

$$c_\gamma(E) = \frac{\varepsilon_{\text{data}}(E)}{\varepsilon_{\text{MC}}(E)}. \tag{14}$$

The calculated efficiencies and correction factors are presented in Table 5. Each entry is shown as central value ± its statistical uncertainty.

Table 5: The photon detection efficiencies in data and MC, and the derived correction factors with their statistical uncertainties for different photon energy intervals. The correction factor uncertainties are calculated using error propagation.

E_γ (GeV)	$\varepsilon_{\text{data}}$	ε_{MC}	c_γ
1.0 – 1.1	0.9744 ± 0.0036	0.9791 ± 0.0027	0.9952 ± 0.0045
1.1 – 1.2	0.9726 ± 0.0038	0.9799 ± 0.0026	0.9926 ± 0.0046
1.2 – 1.3	0.9755 ± 0.0035	0.9798 ± 0.0026	0.9956 ± 0.0044
1.3 – 1.4	0.9724 ± 0.0037	0.9799 ± 0.0026	0.9923 ± 0.0045
1.4 – 1.5	0.9748 ± 0.0035	0.9821 ± 0.0025	0.9926 ± 0.0043
1.5 – 1.6	0.9796 ± 0.0032	0.9873 ± 0.0021	0.9922 ± 0.0039
1.6 – 1.7	0.9690 ± 0.0040	0.9846 ± 0.0024	0.9842 ± 0.0047
≥ 1.8	0.9740 ± 0.0040	0.9824 ± 0.0028	0.9915 ± 0.0049

6.4 PHOKHARA Extra Photon Fraction

A BaBar study of the PHOKHARA generator used for signal simulation found a 20% excess of events with an additional energetic ISR photon along the beam line, relative to data in their $e^+e^- \to \pi^+\pi^-$ process [26]. The ISR-based analysis at BaBar is not affected as it relied on a different generator, but given that our signal process $e^+e^- \to \pi^+\pi^-\pi^0$ is also simulated using PHOKHARA and involves ISR photon emission, this discrepancy warrants a cross-check in our analysis.

To investigate this effect in our channel, we examine the fraction of events with an additional ISR photon directly within our signal process $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\text{ISR}}$. This corresponds to studying events of the type $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\text{ISR}}\gamma_{\text{NLO}}$, where two ISR photons are present in the final state. NNLO contributions are not considered, as their production probability is significantly lower and the additional photons are typically too soft to be reliably identified given our detector resolution and event statistics. Moreover, the limited data sample size does not provide sufficient statistical power to meaningfully extract an NNLO contribution.

In this study, we rely on the ISR photon identified through the 4C kinematic fit procedure described in Sec. 4, which selects the optimal candidate event and tags the leading order (LO) ISR photon with high confidence. Based on this tagged photon, we proceed to investigate the presence of additional photons in the event. Due to the finite coverage of the BESIII electromagnetic calorimeter, the study of NLO photons is divided into two independent categories based on their angular distribution. As this study focuses on the intrinsic discrepancy between data and the PHOKHARA generator, the potential differences between different data-taking rounds are not considered. We perform this investigation using the Round16 dataset and assume the observed effect is representative and can be generalized to the other rounds.

6.4.1 Large Angle NLO Photons

NLO photons emitted at large angles fall within the detector acceptance and can be directly reconstructed from electromagnetic showers. Events containing a large-angle NLO photon are selected by first inverting the nominal signal criterion described in Sec. 4: we require the 4C kinematic fit under the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$ hypothesis to have $\chi^2 > 50$, as events passing the $\chi^2 < 50$ signal selection are, by construction, depleted of additional photons. From this complementary sample, we identify the NLO photon candidate as the extra reconstructed photon that minimizes the χ^2 of a subsequent 4C kinematic fit to the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}\gamma_{\rm NLO}$ hypothesis, which explicitly includes both ISR photons in the final state. The kinematic fit is performed in the laboratory frame, constraining the total four-momentum of all final state particles (π^+, π^-, π^0, the LO ISR photon, and the NLO LA photon) to the initial e^+e^- center-of-mass energy. To ensure a well-reconstructed NLO topology, we further require this 4C fit to satisfy $\chi^2 < 50$. The invariant mass $M(\pi^+\pi^-\pi^0)$ must be within the range $0.7 < M(3\pi) < 2.0$ GeV/c^2.

To suppress background contributions from events where the two ISR photons originate from π^0 or η decays, we examine the invariant mass squared distribution of the LO and NLO LA photon pair, $M^2(\gamma_{\rm ISR}\gamma_{\rm NLO})$. Clear peaks corresponding to the π^0 and η mesons are observed, as shown in Fig. 17. Accordingly, we apply veto requirements of $M^2(\gamma_{\rm ISR}\gamma_{\rm NLO}) < 0.05$ GeV$^2/c^4$ to exclude the π^0 region, and $0.23 < M^2(\gamma_{\rm ISR}\gamma_{\rm NLO}) < 0.33$ GeV$^2/c^4$ to exclude the η region.

To further validate the selection and extract the NLO LA photon yield, we examine the invariant mass squared of the missing system: the total laboratory four-momentum minus the sum of the π^+, π^-, π^0, and LO ISR photon four-momenta. For signal events, the remaining system corresponds exactly to the NLO ISR photon, and this quantity should therefore peak at zero. Figure 18a shows this distribution for events passing the inverted single-ISR selection ($\chi^2 > 50$) with an additional photon candidate.

To extract the signal yield, we perform a fit to this distribution, as shown in Figure 18b:

- The signal shape is taken from simulated $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}\gamma_{\rm NLO}$ events and convolved with a Gaussian to account for possible resolution differences between data and MC. The signal yield is left free in the fit. The Gaussian resolution parameters are extracted from the fit.

- The dominant background from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events is modeled using the shape extracted from simulation. Its yield is constrained using a Gaussian penalty term based on the expected number of events, calculated from the luminosity, the cross-section, and the selection efficiency estimated from MC, with a Poisson error incorporated to account for statistical fluctuations.

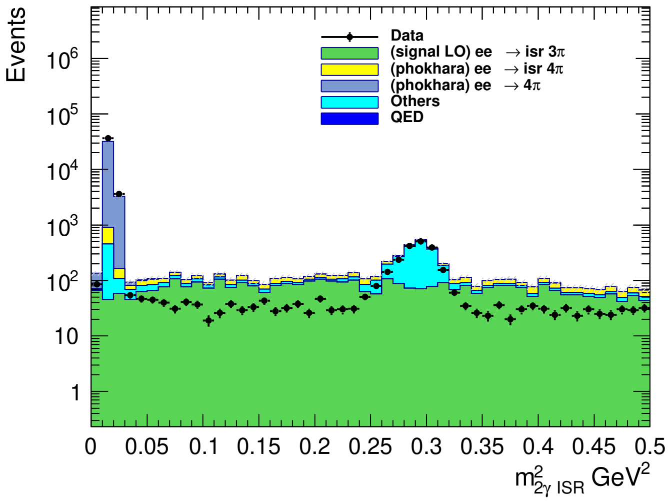


Figure 17: Invariant mass squared distribution of the two ISR photons.

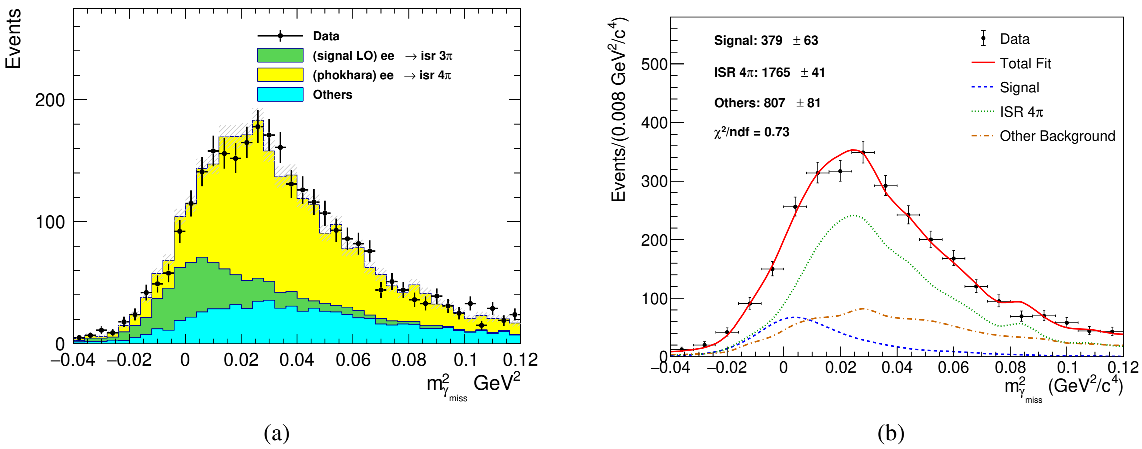


Figure 18: (a) Invariant mass squared of the missing system (total momentum minus $\pi^+\pi^-\pi^0\gamma_{\mathrm{ISR}}$). (b) Fit to the distribution.

- All other remaining background contributions are modeled using shapes extracted from MC samples, with their yields left free in the fit.

The fitted NLO LA signal yield is 379 ± 63. The large uncertainty is due to the low production rate of these events and the significant photon background, which makes it difficult to improve the significance.

6.4.2 Small Angle NLO Photons

NLO photons emitted at very small angles along the beam direction escape detection, and their kinematic properties must be inferred indirectly. The momentum of this undetected photon is calculated from the missing momentum, with the constraint that its transverse components are zero in the center-of-mass frame ($p_x^{\rm CM} = p_y^{\rm CM} = 0$). It is important to note that the beam direction in the laboratory frame is not aligned with the definition of transverse momentum in this context; therefore, we first boost the event to the center-of-mass frame, enforce the transverse momentum condition, and then boost back to the laboratory frame to obtain the four-momentum of the SA photon. The particle is further constrained to have zero mass (photon identity). A 3C kinematic fit is then performed in the laboratory frame, including the detected particles π^+, π^-, π^0, and the LO ISR photon, together with the inferred SA photon, to enforce overall energy and momentum conservation. The invariant mass $M(\pi^+\pi^-\pi^0)$ must be within the range $0.7 < M(3\pi) < 2.0$ GeV/c^2.

Events containing a small-angle NLO photon are selected by first inverting the nominal signal criterion described in Sec. 4: we require the 4C kinematic fit under the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$ hypothesis to have $\chi^2 > 50$. From this complementary sample, we infer the SA photon candidate using the missing momentum technique described above. To ensure a well-constrained NLO topology, we require the 3C kinematic fit to satisfy $\chi^2 < 5$. Additionally, we examine the angular distribution of the inferred SA photon in the center-of-mass frame and require $|\cos\theta_{\gamma_{\rm NLO}}^{\rm CM}| > 0.99$, consistent with the small-angle region where photons escape detection.

To further validate the selection and extract the NLO SA photon yield, we examine the energy distribution of the inferred SA photon, $E_{\gamma_{\rm NLO}}^{\rm lab}$. For signal events, this energy should exhibit a characteristic spectrum determined by the NLO radiation process. Figure 19 shows this distribution for events passing the inverted single-ISR selection and the additional SA photon requirements.

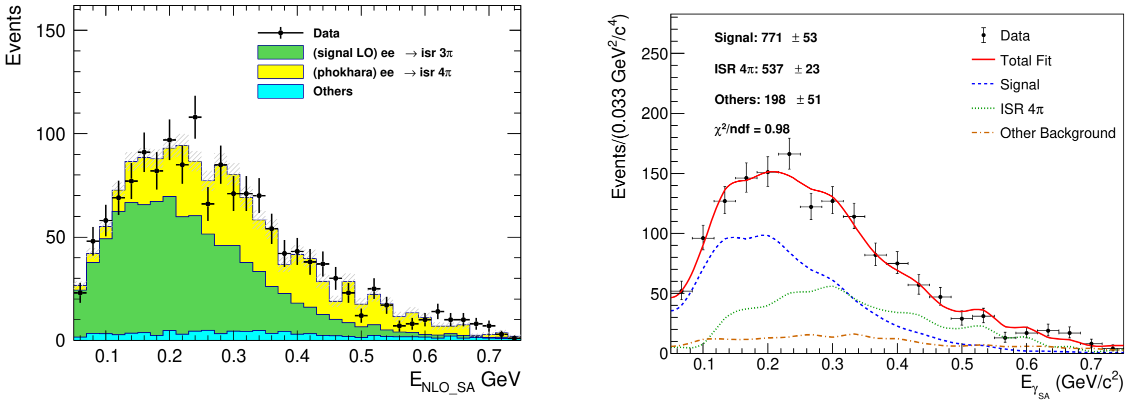
To extract the signal yield, we perform a fit to this distribution, as shown in Figure 19b:

- The signal shape is taken from simulated $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}\gamma_{\rm NLO}$ events and convolved with a Gaussian to account for possible resolution differences between data and MC. The signal yield is left free in the fit. The Gaussian resolution parameters are extracted from the fit.

- The dominant background from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events is modeled using the shape extracted from simulation. Its yield is constrained using a Gaussian penalty term based on the expected number of events, calculated from the luminosity, the cross-section, and the selection efficiency estimated from MC, with a Poisson error incorporated to account for statistical fluctuations.

- All other remaining background contributions are modeled using shapes extracted from MC samples, with their yields left free in the fit.

The fitted NLO SA signal yield is 771 ± 53.



(a) (b)

Figure 19: (a) Laboratory energy distribution of the inferred SA photon candidate. (b) Fit to the distribution.

6.4.3 Data-MC Comparison of NLO Photon Fractions

Following the selection procedures described above for both LA and SA categories, we obtain samples of events containing an identified NLO ISR photon in data. To compare the production rates between data and the PHOKHARA generator, we need to account for the overall selection efficiency that encompasses both detector effects and reconstruction requirements.

Unlike the simplified approach where each truth category is assumed to only pass its corresponding selection, we adopt a more rigorous method that accounts for cross-contamination between categories. Using the PHOKHARA MC sample, we construct an efficiency matrix $\mathcal{A}$ that describes the probability for events of each truth category to pass each of the three selection criteria.

We first classify events at generator level according to the following truth definitions, with an additional requirement on the invariant mass of the three pions consistent with our analysis region:

- The invariant mass $M(\pi^+\pi^-\pi^0)$ must be within the range $0.7 < M(3\pi) < 2.0$ GeV/c^2.
- **LO events**: Events with no NLO photon, or where the NLO photon energy is below 0.05 GeV.
- **NLO LA events**: Events with a NLO photon of energy ≥ 0.05 GeV and $|\cos\theta_{\gamma_{\rm NLO}}| < 0.99$.
- **NLO SA events**: Events with a NLO photon of energy ≥ 0.05 GeV and $|\cos\theta_{\gamma_{\rm NLO}}| > 0.99$.

For each truth category, we determine the number of MC events that pass each of the three reconstruction-level selection criteria: the LO selection (nominal signal selection with 4C fit $\chi^2 < 50$ and a single ISR photon), the NLO LA selection, and the NLO SA selection. The resulting efficiency matrix is presented in Table 6.

The purity of each selection, defined as the fraction of passing events with the correct truth type, is also evaluated. The LO selection purity is 96.34%, while the NLO LA and NLO SA selections achieve purities of 94.35% and 93.11%, respectively. These purities indicate that while each selection is largely dominated by its corresponding truth category, there is non-negligible cross-contamination that must be accounted for in the unfolding.

Table 6: Efficiency matrix $\mathcal{A}$ from PHOKHARA MC. Each entry shows the number of events passing each reconstruction-level selection, with the corresponding efficiency (in percentage) given in parentheses. The "Truth Total" column gives the total number of generated events in each category within the $M(3\pi)$ requirement.

Truth Type	Truth Total	Reco Total	Pass LO	Pass NLO LA	Pass NLO SA
LO	10,437,260	474,675	310,295 (2.9730%)	65 (0.0006%)	260 (0.0025%)
NLO LA	888,289	31,295	3,669 (0.4130%)	2,304 (0.2594%)	385 (0.0433%)
NLO SA	2,250,612	60,327	8,135 (0.3615%)	73 (0.0032%)	8,723 (0.3876%)

The yields in data for each selection category (b_{LO}, $b_{\mathrm{NLO\,LA}}$, $b_{\mathrm{NLO\,SA}}$) are obtained from the fits described in Sections 6.4.1 and 6.4.2 for the NLO LA and SA categories, while the LO yield in data has been previously studied in Fig. 11. These observed yields are related to the unknown generator-level yields x_i (where i denotes LO, NLO LA, or NLO SA) through the efficiency matrix:

$$b_j = \sum_i \mathcal{A}_{ji} x_i,$$

where $\mathcal{A}_{ji}$ is the probability for a truth-type i event to pass selection j. This forms a 3×3 linear system that can be solved for x_i:

$$x = \mathcal{A}^{-1} b.$$

Solving this system using the data yields gives the unfolded generator-level event counts in data, as summarized in Table 7. For comparison, the generator-level fractions from the PHOKHARA MC sample, computed directly from the truth-level classification (including the same $M(3\pi)$ requirement), are also presented alongside the data fractions.

Table 7: Generator-level event yields and fractions in data after unfolding with the efficiency matrix, compared with the truth-level fractions from PHOKHARA MC.

Category	Data		MC Fraction (%)
	Yield	Fraction (%)	
LO	$1,117,269.7 \pm 9,461.1$	77.88 ± 1.45	76.88
NLO LA	$141,352.0 \pm 24,327.7$	9.85 ± 1.57	6.54
NLO SA	$175,919.2 \pm 13,966.1$	12.26 ± 0.92	16.58
Total	$1,434,540.9$	100.00	100.00

Table 8 presents the comparison between data and MC for the three categories. The uncertainties on the unfolded data fractions are propagated from the uncertainties on the fitted yields through the matrix inversion.

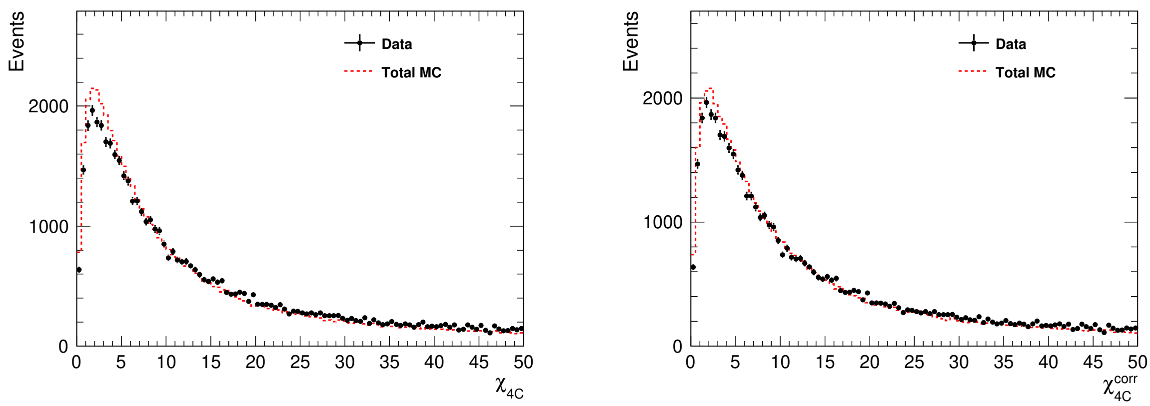
The ratios $f_i^{\mathrm{Data}}/f_i^{\mathrm{MC}}$ reveal a discrepancy between data and the PHOKHARA generator simulation. We observe that PHOKHARA significantly overestimates the production rate of NLO SA events while underestimating NLO LA events. The derived correction factors will be applied in the unfolding procedure through the response matrix to account for this mismodeling of the NLO photon fractions.

Table 8: Comparison of generator-level event fractions between data and PHOKHARA MC after unfolding with the efficiency matrix. The correction factor is defined as the ratio of data to MC fractions and will be applied in the unfolding procedure.

Category	MC Fraction (%)	Data Fraction (%)	Correction factors (Data/MC)
LO	76.88	77.88 ± 1.45	1.013 ± 0.019
NLO LA	6.54	9.85 ± 1.57	1.506 ± 0.240
NLO SA	16.58	12.26 ± 0.92	0.740 ± 0.056

6.5 The χ^2_{4C} distribution and correcting the signal yields

The discrepancy in the χ^2_{4C} distribution between data and MC is corrected following the method established in a dedicated study using $e^+e^- \to K^+K^-\pi^+\pi^-$ events [27]. That analysis, performed under BOSS version 7.1.2, investigated the differences in helix parameters between data and MC simulation across various data-taking periods at the center-of-mass energy of 3.773 GeV. For the signal process $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$, we apply helix parameter corrections to the charged tracks. Figures 20a and 20b show the comparison of χ^2_{4C} distributions between data and MC before and after the correction, respectively. As illustrated, a slight improvement in the agreement between data and MC is observed in the low χ^2_{4C} region, where the majority of events are populated, after applying the helix parameter corrections. With these corrections applied to the signal MC, we subsequently impose a selection requirement of $\chi^2_{4C} < 50$ and fit the signal yields in each $M_{3\pi}$ interval (Sec. 5.2).



(a) Before correction (b) After correction

Figure 20: Comparison of χ^2_{4C} distributions between data and MC for the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$ process (a) before and (b) after applying helix parameter corrections.

6.6 Correcting the signal yields

The migration matrix element A_{ij} is intended to represent the number of events that truly belong to bin j and are reconstructed in bin i. The MC simulation does not perfectly describe data; discrepancies arise from the NLO ISR photon production rate and from the efficiencies of tracking, PID, π^0

reconstruction, and ISR photon detection. These must be corrected before the matrix can be used for unfolding.

Two types of corrections are applied to the MC events. The first correction addresses the discrepancy in the NLO ISR photon production rate. For each event from truth bin j, we assign a weight:

$$w_{\text{NLO}} = c_{\text{NLO}}(j),$$

where $c_{\text{NLO}}(j)$ is the bin-dependent NLO correction factor.

The second correction accounts for the differences in reconstruction efficiencies. For each successfully reconstructed event, we assign an additional weight:

$$w_{\text{eff}} = c_{\text{trk}}(\pi^+) \cdot c_{\text{pid}}(\pi^+) \cdot c_{\text{trk}}(\pi^-) \cdot c_{\text{pid}}(\pi^-) \cdot c_{\pi^0} \cdot c_{\gamma_{\text{ISR}}}.$$

The corrected migration matrix is filled using the following prescription. Initially, all generated events are considered to be lost (i.e., placed into the missing category) with weight w_{NLO}. Then, for each reconstructed event that truly belongs to truth bin j and is observed in reconstructed bin i, we:

- Add $w_{\text{NLO}} \cdot w_{\text{eff}}$ to the migration matrix element A_{ij}^{corr};
- Subtract $w_{\text{NLO}} \cdot w_{\text{eff}}$ from the missing weight to remove the event's contribution from the missing category.

This procedure ensures that the total corrected weight $N_{\text{total},j}^{\text{corr}} = \sum_i A_{ij}^{\text{corr}} + N_{\text{miss},j}^{\text{corr}}$ is conserved. The normalized response matrix P_{ij}^{corr} is then obtained by applying Eq. (8) to A_{ij}^{corr}.

7 Unfolding Procedure

7.1 The IDS unfolding method

To extract the generator-level yield $\left(\frac{dN}{dm}\right)_j^{\text{gen}}$ from the observed yields $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$ and the response matrix P_{ij}, we employ the Iterative Dynamically Stabilized (IDS) unfolding method [24]. This section introduces the mathematical formulation of the method; the specific implementation, parameter optimization, and iteration strategy are described in the following sections.

From the MC simulation, we have constructed the migration matrix A_{ij} (shown in Fig. 12) and the response matrix P_{ij} normalized column-wise as in Eq. (8). The corresponding unfolding matrix is defined row-wise:

$$\tilde{P}_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_u} A_{ik}}, \tag{15}$$

which gives the probability that an event reconstructed in bin i originated from true bin j.

The IDS method uses a regularization function $f(|\Delta|, \tilde{\sigma}, \lambda)$ that dynamically distinguishes statistical fluctuations from genuine signals. Here, $|\Delta|$ represents the absolute difference between data and the model prediction (e.g., $|\Delta d_i|$ for the discrepancy in reconstructed bin i, or $|\Delta u_j|$ for the discrepancy in true bin j), $\tilde{\sigma}$ denotes the corresponding statistical uncertainty ($\tilde{\sigma} d_i$ or $\tilde{\sigma} u_j$), and λ is a tunable regularization parameter. The function satisfies the following asymptotic behavior:

$$\lim_{\frac{|\Delta|}{\lambda\tilde{\sigma}} \to 0} f(|\Delta|, \tilde{\sigma}, \lambda) = 0, \qquad \lim_{\frac{|\Delta|}{\lambda\tilde{\sigma}} \to \infty} f(|\Delta|, \tilde{\sigma}, \lambda) = 1. \tag{16}$$

Thus, for discrepancies much smaller than λ times the uncertainty, the function approaches zero, treating the difference as a fluctuation; for discrepancies much larger, it approaches unity, indicating a real signal that should be unfolded. The transition between these regimes is smooth, avoiding sharp cutoffs that could introduce artifacts.

In the IDS method, the unfolding is performed iteratively. At iteration k, the unfolded spectrum $u_j^{(k)}$ is given by:

$$u_j^{(k)} = \eta \cdot t_j + \sum_{i=1}^{n_d} \left[f\left(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U\right) \Delta d_i^{(k-1)} \tilde{P}_{ij}^{(k-1)} + \left(1 - f\left(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U\right)\right) \Delta d_i^{(k-1)} \delta_{ij} \right], \tag{17}$$

where:

- $u_j^{(k)}$ is the unfolded generator-level yield in bin j at iteration k,
- t_j is the true MC yield in bin j from simulation,
- $\eta = N_d^{\text{MC}}/N_{\text{MC}}$ is the normalization factor that scales the MC to match the data in regions without significant new structures,
- $\Delta d_i^{(k-1)} = d_i - B_i^d - \eta \cdot r_i^{(k-1)}$ is the difference between the observed data yield d_i and the sum of the estimated background B_i^d (which may be zero if no background subtraction is applied) and the normalized reconstructed MC spectrum $\eta \cdot r_i^{(k-1)}$,

- $r_i^{(k-1)}$ is the reconstructed MC spectrum at iteration $k-1$, obtained by folding the unfolded spectrum $u_j^{(k-1)}$ with the response matrix $P_{ij}^{(k-1)}$,
- $\tilde{\sigma} d_i$ is the statistical uncertainty of d_i,
- $\tilde{P}_{ij}^{(k-1)}$ is the unfolding matrix from the previous iteration,
- λ_U is the regularization parameter used in the unfolding step,
- δ_{ij} is the Kronecker delta, which equals 1 if $i = j$ and 0 otherwise.

The first term on the right-hand side, $\eta \cdot t_j$, represents the normalized true MC spectrum, which serves as a baseline. The remaining sum dynamically partitions the discrepancy $\Delta d_i^{(k-1)}$: a fraction f is unfolded using $\tilde{P}_{ij}^{(k-1)}$, while the complementary fraction $1-f$ is retained in the original bin $j = i$ via the Kronecker delta.

The response matrix itself can be iteratively improved. Using the difference $\Delta u_j = u_j - \eta \cdot t_j$ between the current unfolded result and the normalized true MC, the migration matrix is modified as:

$$A'_{ij} = A_{ij} + f\left(|\Delta u_j|, \tilde{\sigma} u_j, \lambda_M\right) \Delta u_j P_{ij} \frac{N_{\mathrm{MC}}}{N_d^{\mathrm{MC}}}, \quad \text{for } i \in \{1, \ldots, n_d\}, \tag{18}$$

where:

- Δu_j is the difference between the current unfolded yield u_j and the normalized true MC $\eta \cdot t_j$,
- $\tilde{\sigma} u_j$ is the statistical uncertainty associated with u_j,
- λ_M is the regularization parameter controlling the modification strength,
- P_{ij} is the current response matrix (column-normalized A_{ij}),
- $N_{\mathrm{MC}} = \sum_j t_j$ is the total number of generated MC events,
- N_d^{MC} is the number of data events compatible with the MC model (used for normalization).

The updated migration matrix A'_{ij} is then renormalized column-wise to obtain an improved response matrix P_{ij}, and row-wise to obtain an improved unfolding matrix $\tilde{P}_{ij}$.

7.2 Parameter optimization and Unfolding

The IDS method involves three regularization parameters: λ_L for the initial unfolding, λ_M for modifying the response matrix, and λ_U for subsequent unfoldings. The iterative procedure consists of the following three steps:

1. **Initial unfolding:** Perform an unfolding using Eq. (17) with $\lambda = \lambda_L$ and the initial response matrix P_{ij}, obtaining $u_j^{(0)}$.
2. **Matrix improvement:** Modify the migration matrix via Eq. (18) using λ_M and the current u_j. Renormalize to obtain updated P_{ij} and $\tilde{P}_{ij}$.

3. **Improved unfolding:** Perform an unfolding using Eq. (17) with $\lambda = \lambda_U$ and the updated matrices.

These three steps can be repeated for a chosen number of iterations.

In this analysis, the four parameters λ_M, λ_L, λ_U, and n_{iter} are optimized by scanning over a grid of possible values. For each parameter combination $(\lambda_M, \lambda_L, \lambda_U, n_{\text{iter}})$, we perform the IDS unfolding and evaluate the quality of the result using a χ^2 figure of merit. The χ^2 is computed by comparing the unfolded spectrum u_j to the true generator-level distribution t_j^{truth} from the MC simulation. The MC sample used for this comparison has been corrected following the procedure described in Sec. 6.6 and is normalized such that its total integral matches the total number of signal events observed in data (shown in Fig. 11), ensuring that the statistical uncertainties from the toy ensembles properly reflect the data statistics.

$$\chi^2 = \left(\mathbf{u} - \mathbf{t}^{\text{truth}}\right)^T \Sigma^{-1} \left(\mathbf{u} - \mathbf{t}^{\text{truth}}\right), \tag{19}$$

where $\mathbf{u}$ and $\mathbf{t}^{\text{truth}}$ are vectors of bin contents, and Σ is the covariance matrix of the unfolded spectrum. The covariance matrix accounts for the statistical uncertainties and their correlations introduced by the unfolding procedure. It is estimated using a MC sampling method:

1. Generate an ensemble of N_{toy} pseudo-data distributions by fluctuating the original observed spectrum according to its statistical uncertainties, using a random number generator.

2. For each pseudo-data distribution, perform the IDS unfolding with the same parameter combination $(\lambda_L, \lambda_U, n_{\text{iter}})$, obtaining an unfolded spectrum $u_j^{(k)}$ for toy k.

3. From the ensemble of unfolded spectra, compute the covariance matrix:

$$\Sigma_{ij} = \frac{1}{N_{\text{toy}} - 1} \sum_{k=1}^{N_{\text{toy}}} \left(u_i^{(k)} - \bar{u}_i\right)\left(u_j^{(k)} - \bar{u}_j\right), \tag{20}$$

where $\bar{u}_i = \frac{1}{N_{\text{toy}}} \sum_k u_i^{(k)}$ is the mean unfolded yield in bin i across the toy ensemble.

In cases where the covariance matrix is singular or ill-conditioned, a small diagonal perturbation $\lambda_{\text{reg}} I$ is added before inversion to ensure numerical stability, with $\lambda_{\text{reg}} = 10^{-6}$ in this analysis.

The optimal parameter combination is chosen based on several considerations: the number of iterations n_{iter} should be as small as possible to avoid unnecessary complexity; the χ^2 value defined in Eq. (19) should be small, indicating good agreement between the unfolded and true spectra; and the χ^2 per degree of freedom, χ^2/n_{dof}, should be close to or less than one, indicating that the uncertainties are well estimated.

It is emphasized that both the χ^2 and the uncertainty estimates are computed strictly within the three intervals and corresponding bin widths defined in Section 5.2. Since reconstructed events near the boundaries may have truth-level counterparts originating from outside the region of interest, we extend the boundaries for each mass region during the training of the unfolding procedure. This extension is applied solely to preserve the completeness of information within the defined region and does not affect the χ^2 or uncertainty calculations, which remain confined to the intervals in Section 5.2. Events falling outside the region of interest are not considered further.

Table 9: The resulting unfolding parameters for the three representative mass regions.

Region	λ_L	λ_U	λ_M	n_{iter}
$\omega(782)$	0.1	0.1	1.0	3
$\phi(1020)$	0.1	0.1	1.0	3
ω'/ω''	0.1	0.1	1.0	3

Using the MC sample from Round 16 as a stand-in for data, a scan is performed over a grid of λ_L, λ_U, and n_{iter} values. The combination that best satisfies the above criteria is selected for the final unfolding of the data. The optimized parameters obtained from this procedure for the three representative mass regions are summarized in Table 9, along with the resulting χ^2/ndf values for all four data-taking rounds to demonstrate consistency.

To validate the stability of the unfolding procedure across different data-taking conditions, the same set of optimized parameters (from Table 9) is applied to the MC samples of other rounds: Round 03/04, Round 15, and Round 17. Figure 21 shows the unfolding results for each round and each mass region.

To further validate the unfolding procedure, we compare the relative statistical uncertainty of the unfolded spectrum with that of the test MC sample. The relative uncertainty of the test MC spectrum, δ_{test}, is computed by summing in quadrature the Poisson errors of each bin and dividing by the total yield:

$$\delta_{\mathrm{test}} = \frac{\sqrt{\sum_i \sigma_i^2}}{\sum_i N_i}, \tag{21}$$

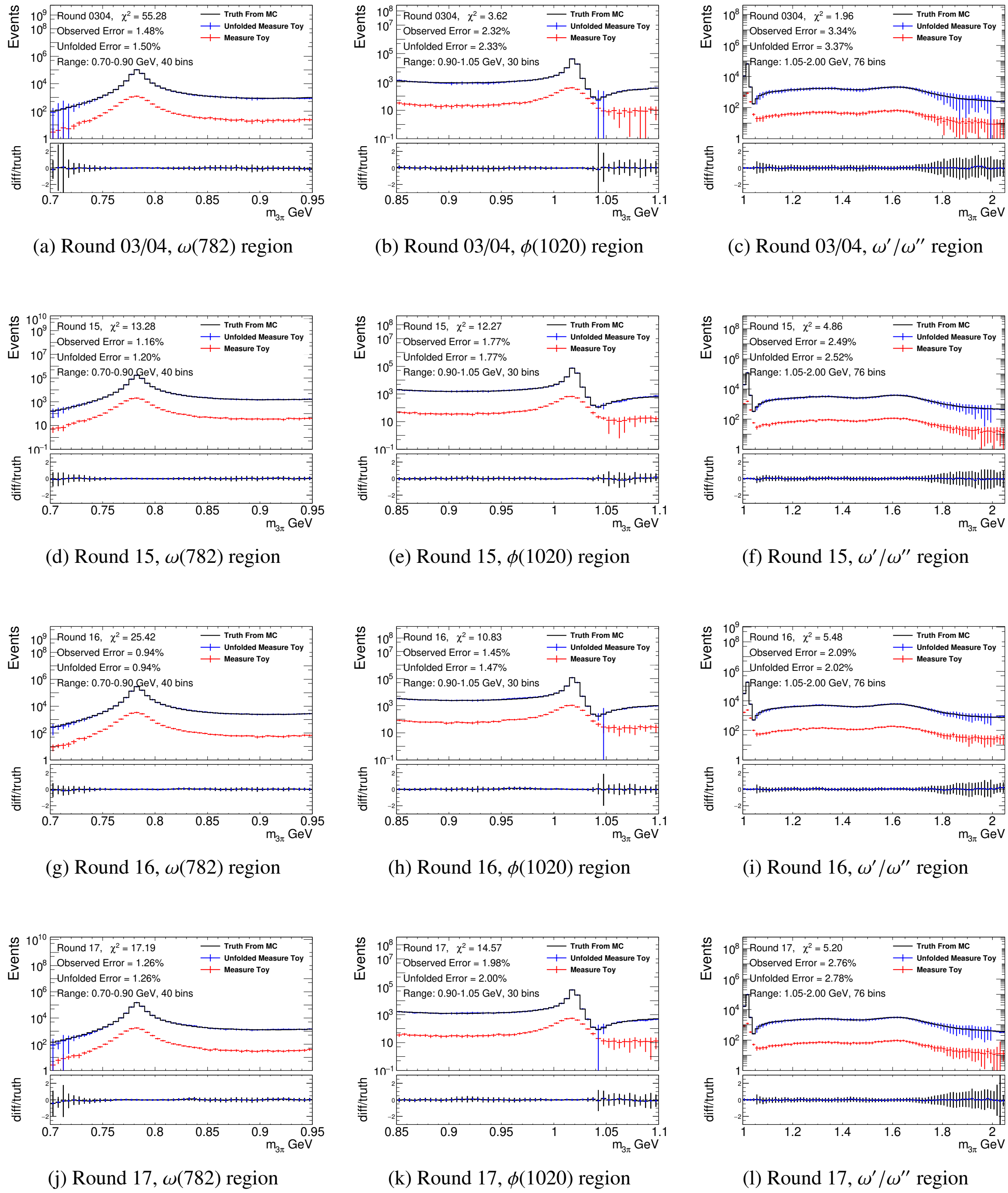
where N_i and σ_i are the content and Poisson error of bin i in the test MC distribution, respectively. This represents the Poisson statistical uncertainty of the test MC sample.

The relative uncertainty of the unfolded spectrum, $\delta_{\mathrm{unfolded}}$, is derived from the full covariance matrix Σ obtained from the toy MC ensemble:

$$\delta_{\mathrm{unfolded}} = \frac{\sqrt{\sum_{i,j} \Sigma_{ij}}}{\sum_i u_i}, \tag{22}$$

where u_i are the bin contents of the unfolded spectrum. The relative uncertainty of the unfolded spectrum should not be smaller than the Poisson uncertainty of the test MC sample.

Using the corrected migration matrix A_{ij}^{corr}, the unfolding of the $M_{3\pi}$ distribution is performed on data from Sec. 5.2. The unfolding procedure automatically accounts for the reconstruction and selection efficiency of the signal by incorporating information from missing events, rather than relying on the response matrix alone. The mass range for each resonance region is extended during the unfolding procedure and subsequently truncated back to the nominal interval defined in Tab. 9 for the final results. Figure 22 presents the unfolded $M_{3\pi}$ distributions for data in the three regions.



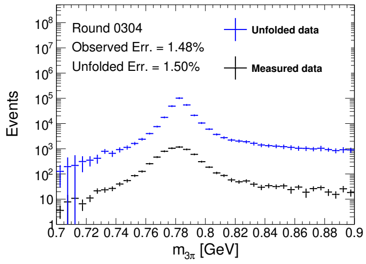
(a) Round 03/04, $\omega(782)$ region (b) Round 03/04, $\phi(1020)$ region (c) Round 03/04, ω'/ω'' region

(d) Round 15, $\omega(782)$ region (e) Round 15, $\phi(1020)$ region (f) Round 15, ω'/ω'' region

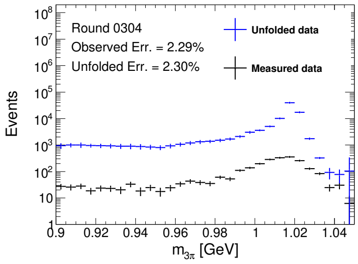
(g) Round 16, $\omega(782)$ region (h) Round 16, $\phi(1020)$ region (i) Round 16, ω'/ω'' region

(j) Round 17, $\omega(782)$ region (k) Round 17, $\phi(1020)$ region (l) Round 17, ω'/ω'' region

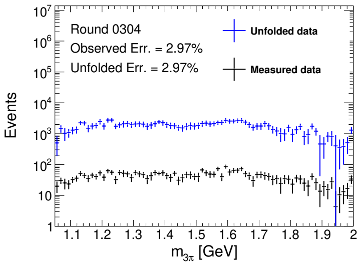
Figure 21: Unfolded results for four data-taking periods (Round 03/04, Round 15, Round 16, Round 17) using the IDS method. The unfolding parameters were optimized using the Round 16 MC sample (third row) and then applied to all rounds.



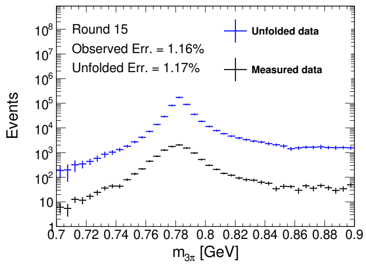
(a) Round 03/04, $\omega(782)$ region



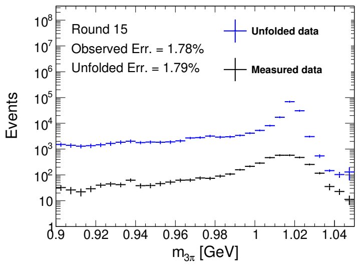
(b) Round 03/04, $\phi(1020)$ region



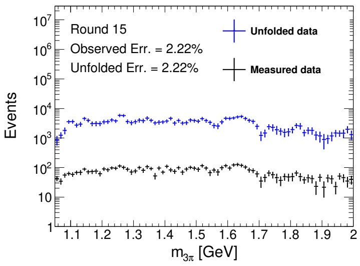
(c) Round 03/04, ω'/ω'' region



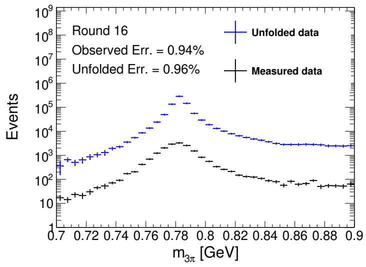
(d) Round 15, $\omega(782)$ region



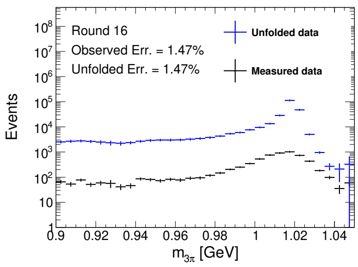
(e) Round 15, $\phi(1020)$ region



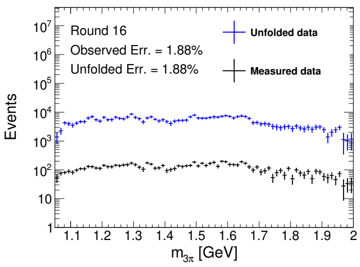
(f) Round 15, ω'/ω'' region



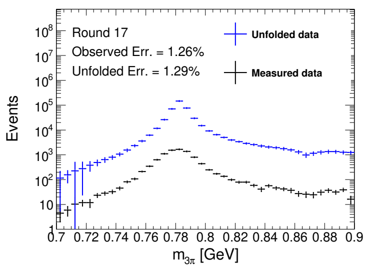
(g) Round 16, $\omega(782)$ region



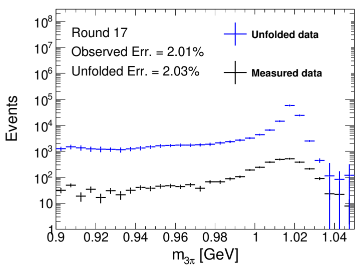
(h) Round 16, $\phi(1020)$ region



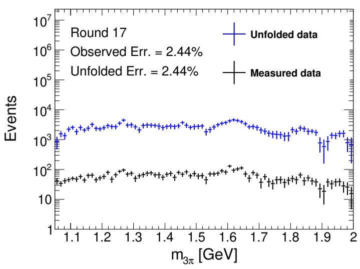
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 22: Unfolded $M_{3\pi}$ distributions after applying the corrected response matrix, shown for four data-taking periods. Each row corresponds to a different round: (a)–(c) Round 03/04, (d)–(f) Round 15, (g)–(i) Round 16, and (j)–(l) Round 17. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right). The results are shown only within the nominal mass ranges defined in Tab. 9.

8 Systematic Uncertainties

In this chapter, we discuss the systematic uncertainties considered in this analysis. The primary systematic uncertainty comes from the efficiency correction applied in the unfolding procedure. Additional uncertainties are associated with the requirement on E/p, unfolding resolution, the 4C kinematic fit, background shape and the π^0 veto.

8.1 Efficiency correction uncertainties

The corrections applied to the MC events in the unfolding procedure are designed to mitigate discrepancies between data and simulation, including the NLO ISR photon production rate and the reconstruction efficiencies for tracking, PID, π^0 reconstruction, and ISR photon detection. The finite precision of these corrections introduces systematic uncertainties in the unfolded results.

The correction factors are grouped into seven independent sources:

$$c_{\mathrm{NLO}}(j),\; c_{\mathrm{trk}}(\pi^+),\; c_{\mathrm{pid}}(\pi^+),\; c_{\mathrm{trk}}(\pi^-),\; c_{\mathrm{pid}}(\pi^-),\; c_{\pi^0},\; c_{\gamma_{\mathrm{ISR}}}.$$

Each factor has an associated uncertainty, derived from auxiliary measurements or theoretical calculations. To propagate these uncertainties to the unfolded spectrum, we perform a toy MC procedure.

For each toy variation $k = 1, \ldots, N_{\mathrm{toy}}$, we independently sample all seven correction factors according to their respective uncertainty distributions (Gaussian with widths given by the factor's uncertainty). This yields a set of seven parameters:

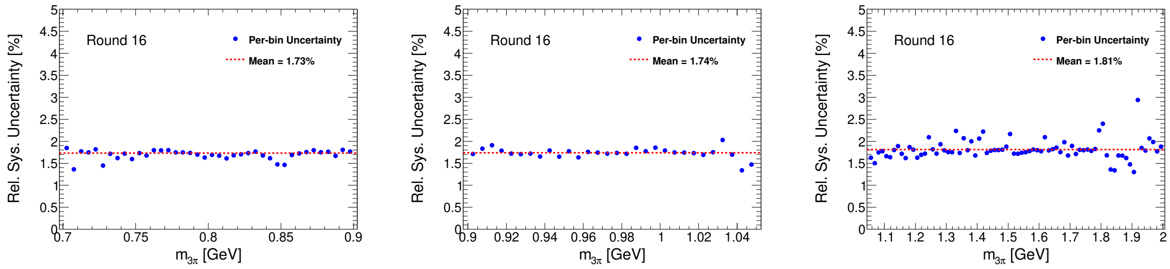
$$(c^{(k)}_{\mathrm{NLO}}(j),\; c^{(k)}_{\mathrm{trk}}(\pi^+),\; c^{(k)}_{\mathrm{pid}}(\pi^+),\; c^{(k)}_{\mathrm{trk}}(\pi^-),\; c^{(k)}_{\mathrm{pid}}(\pi^-),\; c^{(k)}_{\pi^0},\; c^{(k)}_{\gamma_{\mathrm{ISR}}}).$$

For each such parameter set, we recompute the corrected migration matrix $A^{\mathrm{corr},(k)}_{ij}$ following the prescription in Sec. 6.6, then derive the normalized response matrix and perform the unfolding to obtain a toy unfolded spectrum $\{u^{(k)}_i\}$, where i indexes the truth bins.

After processing all N_{toy} toys, the covariance matrix of the systematic uncertainty arising from the efficiency corrections is estimated as:

$$\Sigma^{\mathrm{sys}}_{ij} = \frac{1}{N_{\mathrm{toy}} - 1} \sum_{k=1}^{N_{\mathrm{toy}}} \left(u^{(k)}_i - u^{\mathrm{nom}}_i\right)\left(u^{(k)}_j - u^{\mathrm{nom}}_j\right),$$

where u^{nom}_i is the nominal unfolded spectrum. The diagonal elements $\sqrt{\Sigma^{\mathrm{sys}}_{ii}}$ provide the uncertainty per bin, while the off-diagonal elements quantify correlations between bins induced by the uncertainties in the correction factors. This covariance matrix is taken as the systematic uncertainty contribution from the efficiency corrections in the final results. This procedure is performed independently for each data-taking period (rounds 0304, 15, 16, and 17) and for each $M_{3\pi}$ region. The resulting diagonal systematic uncertainties per bin, expressed as relative uncertainties, are shown in Fig. 23 for all regions of round 16.



(a) Round 16, $\omega(782)$ region (b) Round 16, $\phi(1020)$ region (c) Round 16, ω'/ω'' region

Figure 23: Diagonal relative systematic uncertainties (per bin) from efficiency corrections, shown for Round 16. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right). For each bin, the relative uncertainty is computed as $\sqrt{\Sigma_{ii}^{\text{sys}}}/u_i$, where u_i is the unfolded yield.

8.2 The detector resolution effects in unfolding

The migration of events between neighboring mass bins due to detector resolution is quantified by the response matrix P_{ij} constructed from MC simulations. As an example, Fig. 24 shows the distribution of reconstructed $m_{3\pi}$ for events generated in the truth bin $[0.785, 0.790]$ GeV$/c^2$ using the round 16 MC sample. A clear broadening is observed around the ω meson peak, which reflects the detector resolution as modeled in the MC simulation.

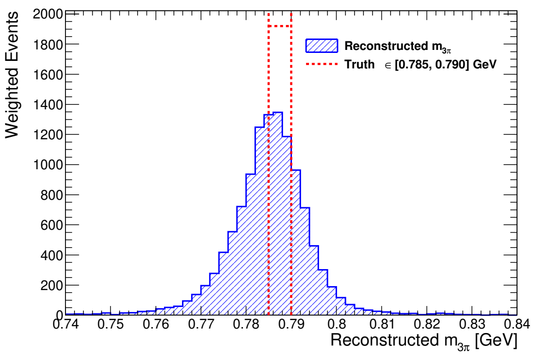


Figure 24: Distribution of reconstructed $m_{3\pi}$ for MC events generated in the truth bin $[0.785, 0.790]$ GeV$/c^2$ (round 16). The spread illustrates the detector resolution effect as modeled in MC, which causes migrations to neighboring bins.

The observed yield in each mass bin is related to the generator-level yield through the response matrix:

$$\left(\frac{dN}{dm}\right)_i^{\text{obs}} = \sum_j P_{ij} \left(\frac{dN}{dm}\right)_j^{\text{gen}} \tag{23}$$

where P_{ij} is constructed entirely from MC simulations. Both data and MC events suffer from detector resolution broadening, but the response matrix only contains the MC-predicted migration pattern. A key concern is whether the detector resolution in data differs from that in MC, as any mismatch would bias the unfolded cross section.

To quantify this potential discrepancy, we introduce a calibrated response matrix $P_{ij}^{\text{cal}}(\delta,\kappa)$ that parameterizes the differences between data and MC. For each truth bin j, the mean and standard deviation of the reconstructed mass distribution from MC are defined as:

$$\mu_{0,j} = \frac{\sum_i m_i P_{ij}}{\sum_i P_{ij}}, \qquad \sigma_{0,j} = \sqrt{\frac{\sum_i (m_i - \mu_{0,j})^2 P_{ij}}{\sum_i P_{ij}}} \tag{24}$$

where m_i is the center of reconstructed bin i. The data response is then modeled by applying a global shift δ and a broadening factor κ to these MC-derived parameters:

$$\mu_j = \mu_{0,j} + \delta, \qquad \sigma_j = \kappa\sigma_{0,j} \tag{25}$$

These calibrated parameters define a Gaussian smearing kernel $G_{ik}^{(j)}(\delta,\kappa)$ for each truth bin j, which is then convolved with the original MC response matrix to obtain the calibrated transfer matrix:

$$P_{ij}^{\text{cal}}(\delta,\kappa) = \sum_k G_{ik}^{(j)}(\delta,\kappa) P_{kj}^{(0)} \tag{26}$$

where $P_{kj}^{(0)}$ is the nominal MC response matrix. The observed data distribution is then modeled as:

$$\left(\frac{dN}{dm}\right)_i^{\text{obs}} = \sum_j P_{ij}^{\text{cal}}(\delta,\kappa) \left(\frac{dN}{dm}\right)_j^{\text{VMD}} \tag{27}$$

By fitting Eq. (27) to the observed data using the known VMD lineshape as the true spectrum [30, 19], the fitted parameters δ and κ directly quantify the shift and broadening differences between data and MC in the response matrix. Both fitted values are found to be well below the bin width, rendering the associated systematic uncertainty from detector resolution effects negligible.

8.3 The requirement on E/p

To evaluate the systematic uncertainty associated with the $E/p < 0.8$ requirement in the $e^+e^- \to \pi^+\pi^-\pi^0$ process, we adopt a data-driven approach using the control sample of $e^+e^- \to \pi^+\pi^-\pi^0$ near the J/ψ mass following Ref. [29]. Based on this work, we study the efficiency discrepancy between data and MC as a function of pion momentum, treating π^+ and π^- independently due to possible detector asymmetries. The momentum range is divided into eight bins: < 0.2, $0.2 - 0.4$, $0.4 - 0.6$, $0.6 - 0.8$, $0.8 - 1.0$, $1.0 - 1.2$, $1.2 - 1.4$, and > 1.4 GeV/c.

The efficiency of the $E/p < 0.8$ cut in a given momentum bin is defined as:

$$\varepsilon(p) = \frac{N_{\text{pass}}(p)}{N_{\text{total}}(p)},$$

where $N_{\text{total}}(p)$ is the number of events before the E/p cut, and $N_{\text{pass}}(p)$ is the number after the cut. We compute this efficiency separately for data using the J/ψ control sample and for signal MC. The efficiency ratio $R(p) = \varepsilon_{\text{data}}(p)/\varepsilon_{\text{MC}}(p)$ serves as a correction factor.

The results for π^+ (studying $E/p_{\pi^+} < 0.8$ with $E/p_{\pi^-} < 0.8$ fixed) are shown in Table 10. The results for π^- (studying $E/p_{\pi^-} < 0.8$ with $E/p_{\pi^+} < 0.8$ fixed) are shown in Table 11.

Table 10: Efficiency of the $E/p_{\pi^+} < 0.8$ cut as a function of π^+ momentum, with $E/p_{\pi^-} < 0.8$ fixed. The efficiency ratio $R = \varepsilon_{\mathrm{data}}/\varepsilon_{\mathrm{MC}}$ and its uncertainty are shown.

p_{π^+} [GeV/c]	$\varepsilon_{\mathrm{data}}$ [%]	$\varepsilon_{\mathrm{MC}}$ [%]	R
< 0.2	99.63 ± 0.74	99.69 ± 0.94	0.9994 ± 0.0147
$0.2 - 0.4$	89.52 ± 0.27	88.79 ± 0.32	1.0082 ± 0.0061
$0.4 - 0.6$	90.37 ± 0.19	91.85 ± 0.23	0.9839 ± 0.0039
$0.6 - 0.8$	94.62 ± 0.17	96.57 ± 0.20	0.9798 ± 0.0033
$0.8 - 1.0$	96.18 ± 0.17	97.78 ± 0.20	0.9836 ± 0.0032
$1.0 - 1.2$	97.08 ± 0.20	98.19 ± 0.22	0.9887 ± 0.0037
$1.2 - 1.4$	97.58 ± 0.20	98.55 ± 0.24	0.9902 ± 0.0039
> 1.4	93.50 ± 0.14	98.85 ± 0.17	0.9459 ± 0.0025

Table 11: Efficiency of the $E/p_{\pi^-} < 0.8$ cut as a function of π^- momentum, with $E/p_{\pi^+} < 0.8$ fixed. The efficiency ratio $R = \varepsilon_{\mathrm{data}}/\varepsilon_{\mathrm{MC}}$ and its uncertainty are shown.

p_{π^-} [GeV/c]	$\varepsilon_{\mathrm{data}}$ [%]	$\varepsilon_{\mathrm{MC}}$ [%]	R
< 0.2	98.96 ± 0.74	98.67 ± 0.94	1.0029 ± 0.0147
$0.2 - 0.4$	93.32 ± 0.28	91.44 ± 0.33	1.0206 ± 0.0062
$0.4 - 0.6$	97.39 ± 0.20	96.73 ± 0.23	1.0068 ± 0.0041
$0.6 - 0.8$	97.91 ± 0.17	97.81 ± 0.20	1.0010 ± 0.0035
$0.8 - 1.0$	98.42 ± 0.17	98.31 ± 0.20	1.0011 ± 0.0035
$1.0 - 1.2$	98.66 ± 0.20	98.54 ± 0.22	1.0012 ± 0.0040
$1.2 - 1.4$	98.61 ± 0.20	98.87 ± 0.24	0.9974 ± 0.0042
> 1.4	94.54 ± 0.15	99.09 ± 0.17	0.9541 ± 0.0026

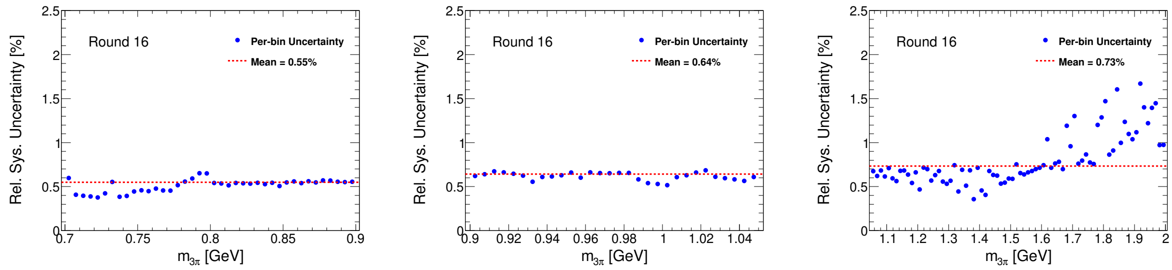
To propagate the uncertainty of the E/p requirement to the cross section measurement, we follow the same toy MC procedure described in Sec. 8.1. The efficiency ratio $R(p)$ is treated as an additional correction factor applied to the MC events. For each toy variation, the combined correction factor for an event is:

$$c_{E/p}^{(k)} = \sqrt{R_{\pi^+}^{(k)}(p_{\pi^+}) \cdot R_{\pi^-}^{(k)}(p_{\pi^-})},$$

where $R_{\pi^+}^{(k)}(p_{\pi^+})$ and $R_{\pi^-}^{(k)}(p_{\pi^-})$ are sampled independently for each momentum bin according to Gaussian distributions. After processing all N_{toy} toys, the covariance matrix of the systematic uncertainty arising from the E/p correction is estimated as:

$$\Sigma^{\text{sys}}_{ij} = \frac{1}{N_{\text{toy}} - 1} \sum_{k=1}^{N_{\text{toy}}} \left(u_i^{(k)} - u_i^{\text{nom}}\right)\left(u_j^{(k)} - u_j^{\text{nom}}\right),$$

where u_i^{nom} is the nominal unfolded spectrum. The diagonal elements $\sqrt{\Sigma^{\text{sys}}_{ii}}$ provide the uncertainty per bin, while the off-diagonal elements quantify correlations between bins induced by the uncertainties in the E/p efficiency ratio. This covariance matrix is taken as the systematic uncertainty contribution from the E/p requirement in the final results. This procedure is performed independently for each data-taking period (rounds 0304, 15, 16, and 17) and for each $M_{3\pi}$ region. The resulting diagonal systematic uncertainties per bin, expressed as relative uncertainties, are shown in Fig. 25 for all regions of round 16.



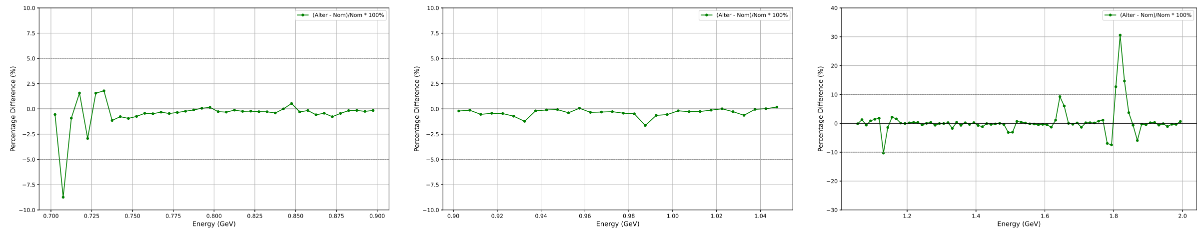
(a) Round 16, $\omega(782)$ region (b) Round 16, $\phi(1020)$ region (c) Round 16, ω'/ω'' region

Figure 25: Diagonal relative systematic uncertainties (per bin) from the E/p requirement, shown for Round 16. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right). For each bin, the relative uncertainty is computed as $\sqrt{\Sigma^{\text{sys}}_{ii}}/u_i$, where u_i is the unfolded yield.

8.4 The 4C kinematic fit

The nominal results for both the measured spectrum $\left(\frac{dN}{dm}\right)$ and the response matrix used in the unfolding procedure are obtained with the $\chi^2_{\text{corr}} < 50$ requirement applied in the 4C kinematic fit. To evaluate the associated systematic uncertainty, we repeat the entire analysis chain — including the estimation of $\left(\frac{dN}{dm}\right)$ and the unfolding procedure — after disabling the correction of the helix parameters for charged tracks in the kinematic fit. The differences between the alternative and nominal unfolded spectra, defined as $\Delta_i = \left(\frac{dN}{dm}\right)^{\text{alt}}_i - \left(\frac{dN}{dm}\right)^{\text{nom}}_i$, are shown in Fig. 26 for all three $M_{3\pi}$ regions of round 16.

In principle, one could vary the helix parameter corrections according to their uncertainties and repeat the unfolding multiple times to extract the covariance matrix, as described in Sec. 8.1. However, the uncertainties on these corrections are very small, and such an approach would likely lead to an underestimate of the systematic uncertainty. To be conservative, we instead take half of the difference between the unfolded spectra obtained with the alternative and nominal configurations as the systematic uncertainty for each bin. To account for bin-to-bin correlations, we construct the full covariance matrix as described below.



(a) Round 16, $\omega(782)$ region (b) Round 16, $\phi(1020)$ region (c) Round 16, ω'/ω'' region

Figure 26: Differences Δ_i between the alternative and nominal unfolded spectra for Round 16.

Let Δ_i be the difference between the alternative and nominal unfolded spectra in bin i:

$$\Delta_i = \left(\frac{dN}{dm}\right)_i^{\text{alt}} - \left(\frac{dN}{dm}\right)_i^{\text{nom}} \tag{28}$$

The diagonal elements of the covariance matrix are given by:

$$\text{Cov}_{ii} = \left(\frac{1}{2}\Delta_i\right)^2 \tag{29}$$

The off-diagonal elements are constructed as:

$$\text{Cov}_{ij} = s_i \cdot s_j \cdot \sqrt{\text{Cov}_{ii} \cdot \text{Cov}_{jj}},$$

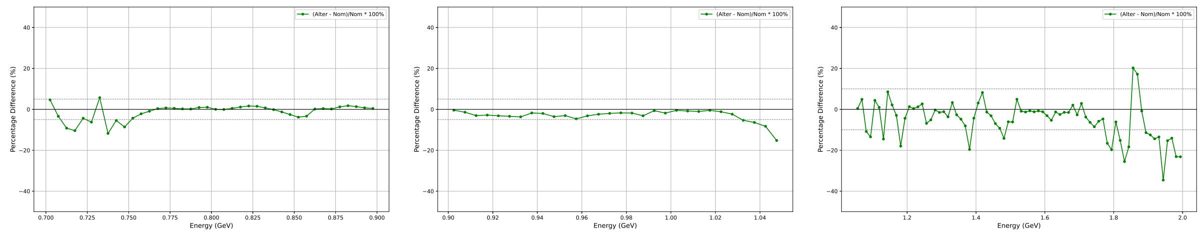
where $s_i = \text{sign}(\Delta_i)$.

This procedure yields a complete covariance matrix that accounts for bin-to-bin correlations: bins with the same sign are assumed to be fully positively correlated (+1), while bins with opposite signs are fully anti-correlated (−1). The resulting covariance matrix is then added in quadrature to the total covariance matrix of the cross section measurement.

8.5 Background estimate

The background in the $\pi^+\pi^-\pi^0$ sample consists mainly of two components. The dominant background arises from the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ process. Its contribution is estimated using a data-driven method, where the background level in each 3π mass bin is evaluated and accompanied by a statistical uncertainty. In the fit to extract the 3π yield, this background component is constrained via a Gaussian penalty term that incorporates its uncertainty. Therefore, no additional systematic uncertainty is assigned for this dominant background.

For all other remaining background processes, a first- or second-order polynomial is used in the nominal fit, with all parameters left floating. The main systematic uncertainty associated with these backgrounds comes from their shape. To evaluate this effect, we perform an alternative fit in which the background shape is taken from the inclusive MC sample, and the normalization is constrained via a Gaussian term fixed to the expected yield based on the integrated luminosity. The resulting 3π yield after the alternative fit is then propagated through the full unfolding procedure. The differences between the alternative and nominal unfolded spectra, defined as $\Delta_i = \left(\frac{dN}{dm}\right)_i^{\text{alt}} - \left(\frac{dN}{dm}\right)_i^{\text{nom}}$, are shown in Fig. 27 for all three $M_{3\pi}$ regions of round 16.



(a) Round 16, $\omega(782)$ region (b) Round 16, $\phi(1020)$ region (c) Round 16, ω'/ω'' region

Figure 27: Differences Δ_i between the alternative and nominal unfolded spectra for Round 16.

Following the same procedure described in Sec. 8.4, we construct a covariance matrix using Δ_i (the difference between the alternative and nominal unfolded spectra) instead of $\frac{1}{2}\Delta_i$ for the diagonal elements. The diagonal elements of the covariance matrix are given by:

$$\text{Cov}_{ii} = \Delta_i^2,$$

The off-diagonal elements are constructed as:

$$\text{Cov}_{ij} = s_i \cdot s_j \cdot \sqrt{\text{Cov}_{ii} \cdot \text{Cov}_{jj}},$$

where $s_i = \text{sign}(\Delta_i)$. This covariance matrix is then added in quadrature to the total covariance matrix of the cross section measurement.

8.6 The π^0 veto

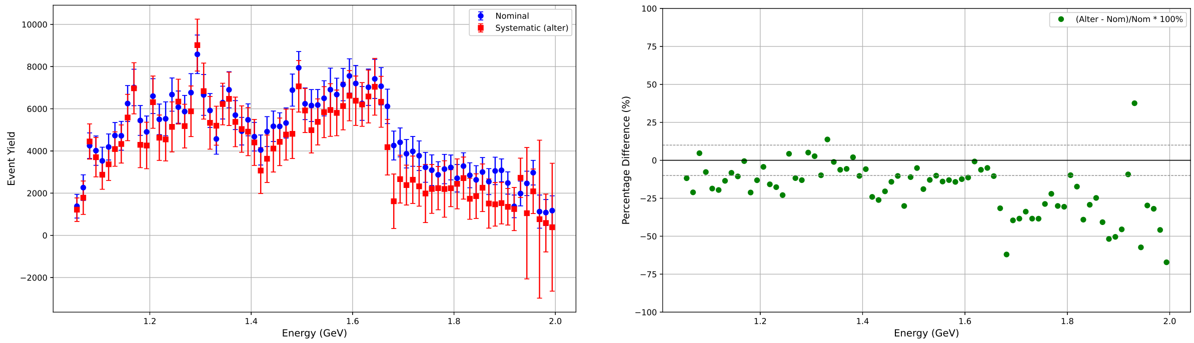
The π^0 veto condition is applied only in the high-mass region of the spectrum. To evaluate the associated systematic uncertainty, we repeat the analysis — including the unfolding procedure — after removing this veto condition, focusing specifically on the third mass interval where the veto is originally applied. The comparison between the unfolded spectra obtained with and without the π^0 veto, as well as their relative differences $\Delta_i = \left(\frac{dN}{dm}\right)_i^{\text{no veto}} - \left(\frac{dN}{dm}\right)_i^{\text{veto}}$, are shown in Fig. 28 for the high-mass region of round 16.

Following the same procedure described in Sec. 8.4, we construct a covariance matrix using Δ_i (the difference between the unfolded spectra obtained without and with the π^0 veto) for the diagonal elements:

$$\text{Cov}_{ii} = \Delta_i^2,$$

The off-diagonal elements are constructed as:

$$\text{Cov}_{ij} = s_i \cdot s_j \cdot \sqrt{\text{Cov}_{ii} \cdot \text{Cov}_{jj}},$$



(a) Comparison of unfolded spectra with and without π^0 veto

(b) Relative differences

Figure 28: Systematic uncertainty from the π^0 veto in the high-mass region of Round 16. The left panel shows the unfolded spectra obtained with (nominal) and without (alternative) the π^0 veto condition. The right panel shows the relative differences $\Delta_i/u_i^{\text{nom}}$, where $\Delta_i = u_i^{\text{no veto}} - u_i^{\text{veto}}$.

where $s_i = \text{sign}(\Delta_i)$. This covariance matrix is then added in quadrature to the total covariance matrix of the cross section measurement.

9 Summary

9.1 Calculation of cross sections

From the unfolding procedure, we obtain the unfolded $M_{3\pi}$ spectrum $\left(\frac{dN}{dm}\right)_i^{\text{unfold}}$ and its covariance matrix $\mathbf{C}$ in discrete mass bins. Here, $\mathbf{C}$ denotes the total covariance matrix of the unfolded spectrum, which includes contributions from both statistical uncertainties and all systematic uncertainties studied in Sec. 8. The diagonal elements $\sqrt{[\mathbf{C}]_{ii}}$ give the total uncertainty Δy_i of each bin, i.e., $\Delta y_i = \sqrt{[\mathbf{C}]_{ii}}$.

The dressed cross section in bin i is given by:

$$\sigma_i^{\text{dress}} = \frac{\left(\frac{dN}{dm}\right)_i^{\text{unfold}}}{\Delta L_i \cdot \mathcal{B}(\pi^0 \to \gamma\gamma)}, \tag{30}$$

where ΔL_i is the integrated luminosity in that bin, and $\mathcal{B}(\pi^0 \to \gamma\gamma) = (98.823 \pm 0.034)\%$ is the branching fraction from the PDG [28]. The calculation of $\frac{dL}{dm}$ is described in Sec. 5.

To properly propagate uncertainties, we first define an intermediate cross section by normalizing only to the integrated luminosity:

$$\sigma_i^{\text{interm}} = \frac{\left(\frac{dN}{dm}\right)_i^{\text{unfold}}}{\Delta L_i}. \tag{31}$$

Its covariance matrix combines the unfolded uncertainty and the fully correlated luminosity uncertainty:

$$\left[\mathbf{C}^{(\text{interm})}\right]_{ij} = \frac{[\mathbf{C}]_{ij}}{\Delta L_i \Delta L_j} + \delta_L^2 \cdot \sigma_i^{\text{interm}} \sigma_j^{\text{interm}}, \tag{32}$$

where δ_L is the relative uncertainty of luminosity listed in Table 1.

The dressed cross section is then obtained by dividing by $\mathcal{B}$:

$$\sigma_i^{\text{dress}} = \frac{\sigma_i^{\text{interm}}}{\mathcal{B}}. \tag{33}$$

Propagating the fully correlated branching fraction uncertainty ($\delta_{\mathcal{B}} = 0.034/98.823 \approx 0.034\%$) gives the final covariance matrix:

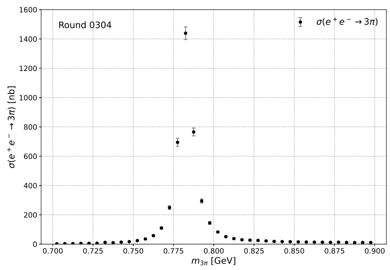
$$\left[\mathbf{C}^{(\text{dress})}\right]_{ij} = \frac{\left[\mathbf{C}^{(\text{interm})}\right]_{ij}}{\mathcal{B}^2} + \delta_{\mathcal{B}}^2 \cdot \sigma_i^{\text{dress}} \sigma_j^{\text{dress}}. \tag{34}$$

For a single bin, the total relative uncertainty adds in quadrature:

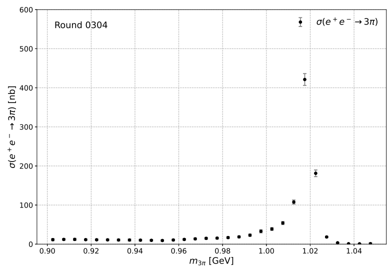
$$\left(\frac{\Delta\sigma_i^{\text{dress}}}{\sigma_i^{\text{dress}}}\right)^2 = \left(\frac{\Delta y_i}{y_i}\right)^2 + \delta_L^2 + \delta_{\mathcal{B}}^2, \tag{35}$$

where $\Delta y_i / y_i = \sqrt{[\mathbf{C}]_{ii}} / \left(\frac{dN}{dm}\right)_i^{\text{unfold}}$ is the total relative uncertainty from unfolding, combining both statistical and all systematic sources.

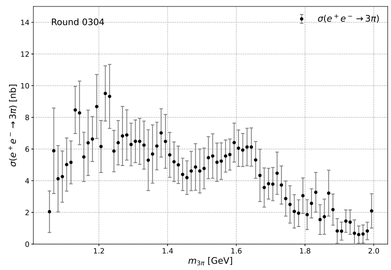
Using Eq. 30 - Eq. 34, we obtain four independent dressed cross section results for Round 03/04, Round 15, Round 16, and Round 17, shown in Fig. 29.



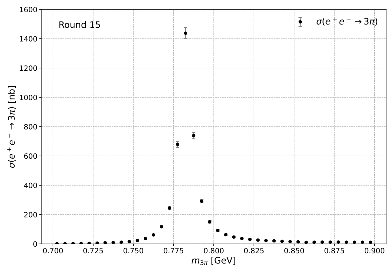
(a) Round 03/04, $\omega(782)$ region



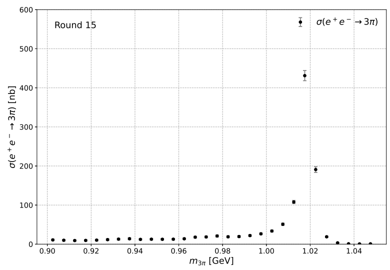
(b) Round 03/04, $\phi(1020)$ region



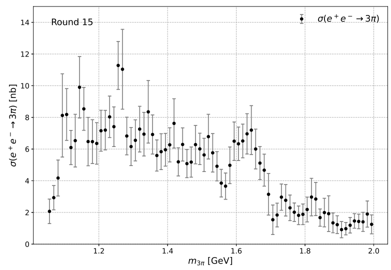
(c) Round 03/04, ω'/ω'' region



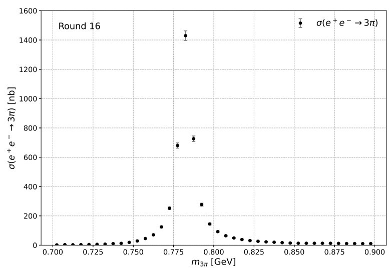
(d) Round 15, $\omega(782)$ region



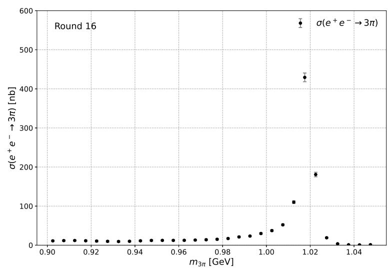
(e) Round 15, $\phi(1020)$ region



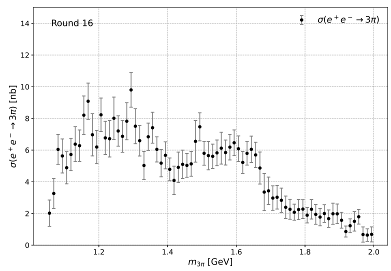
(f) Round 15, ω'/ω'' region



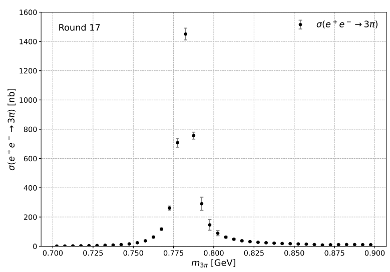
(g) Round 16, $\omega(782)$ region



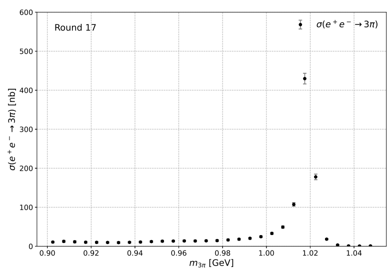
(h) Round 16, $\phi(1020)$ region



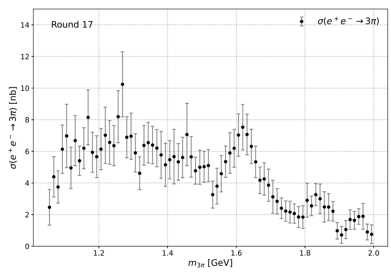
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 29: Dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$ results for the four independent data-taking periods (Round 03/04, Round 15, Round 16, and Round 17). The uncertainties are derived from the covariance matrices obtained from the unfolding procedure, which fully describe the correlations of the unfolded mass spectrum.

9.2 Combination of independent dressed cross section measurements

Since the four data-taking periods (Round 03/04, Round 15, Round 16, and Round 17) are statistically independent, we can combine their dressed cross section results to obtain a more precise measurement. However, a simple weighted average using only the diagonal uncertainties would be suboptimal, as it neglects the bin-to-bin correlations within each measurement that arise from the unfolding procedure.

To properly account for these correlations, we employ the Best Linear Unbiased Estimator (BLUE) method [31]. BLUE provides a systematic framework for combining multiple correlated measurements that estimate the same underlying quantity, producing an estimate that is both unbiased and has minimum variance.

9.2.1 Formalism of BLUE

Suppose we have n independent measurements $\mathbf{x}_i$ ($i = 1, \ldots, n$), each being a vector of length m (corresponding to the m invariant mass bins). Each measurement is an unbiased estimator of the true value $\boldsymbol{\mu}$:

$$E[\mathbf{x}_i] = \boldsymbol{\mu}, \quad \mathrm{Cov}(\mathbf{x}_i) = \mathbf{V}_i \tag{36}$$

where $\mathbf{V}_i$ is the $m \times m$ covariance matrix for the i-th measurement, which contains both the variances ($\sigma_{i,p}^2$ on the diagonal) and the bin-to-bin covariances ($\rho_{pq}\sigma_{i,p}\sigma_{i,q}$ off-diagonal). Since the measurements are independent, the cross-covariances satisfy $\mathrm{Cov}(\mathbf{x}_i, \mathbf{x}_j) = 0$ for $i \neq j$.

We seek a linear combination of all measurements:

$$\hat{\boldsymbol{\mu}} = \sum_{i=1}^{n} \mathbf{W}_i \mathbf{x}_i \tag{37}$$

where each $\mathbf{W}_i$ is an $m \times m$ weight matrix to be determined. The estimator $\hat{\boldsymbol{\mu}}$ is required to be unbiased:

$$E[\hat{\boldsymbol{\mu}}] = \sum_{i=1}^{n} \mathbf{W}_i E[\mathbf{x}_i] = \left(\sum_{i=1}^{n} \mathbf{W}_i\right)\boldsymbol{\mu} = \boldsymbol{\mu} \tag{38}$$

which imposes the constraint $\sum_{i=1}^{n} \mathbf{W}_i = \mathbf{I}$ (the identity matrix).

The covariance matrix of the combined estimator is:

$$\mathbf{U} = \mathrm{Cov}(\hat{\boldsymbol{\mu}}) = \sum_{i=1}^{n} \mathbf{W}_i \mathbf{V}_i \mathbf{W}_i^T \tag{39}$$

since the cross-terms vanish due to independence.

9.2.2 Optimal weights

The BLUE method minimizes the variance (in the sense of matrix trace or any positive-definite metric) subject to the unbiasedness constraint. Solving this optimization problem using Lagrange multipliers yields the optimal weight matrices:

$$\mathbf{W}_i = \left(\sum_{j=1}^{n} \mathbf{V}_j^{-1} \right)^{-1} \mathbf{V}_i^{-1} \tag{40}$$

provided that all $\mathbf{V}_i$ are invertible. If some covariance matrices are singular or near-singular, we use the Moore-Penrose pseudo-inverse instead.

Substituting these weights back gives the combined estimate:

$$\hat{\boldsymbol{\mu}} = \left(\sum_{i=1}^{n} \mathbf{V}_i^{-1} \right)^{-1} \left(\sum_{i=1}^{n} \mathbf{V}_i^{-1} \mathbf{x}_i \right) \tag{41}$$

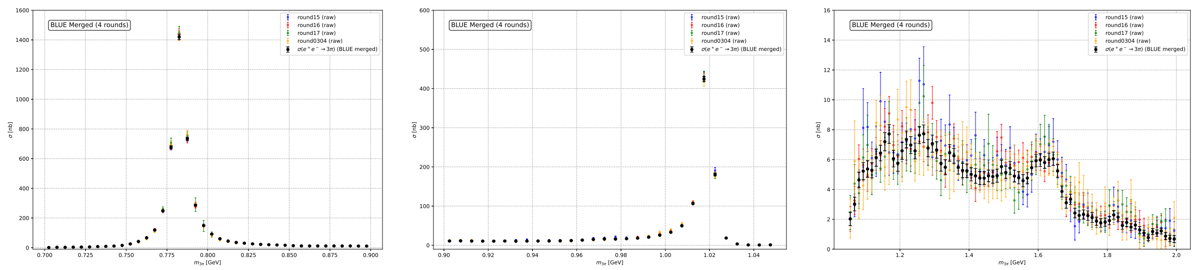
and its covariance matrix:

$$\mathbf{U} = \left(\sum_{i=1}^{n} \mathbf{V}_i^{-1} \right)^{-1} \tag{42}$$

9.2.3 Application to this analysis

We apply the BLUE method to combine the four independent measurements of the dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$. For each data-taking period, the measurement is represented by a vector $\mathbf{x}_i$ (size 40 for Region 1, 30 for Region 2, and 76 for Region 3) and its associated covariance matrix $\mathbf{V}_i$, which accounts for both statistical uncertainties and bin-to-bin correlations from the unfolding procedure. The combined result $\hat{\boldsymbol{\mu}}$ represents the best estimate of $\sigma_{3\pi}^{\text{dress}}(m)$, with its covariance matrix $\mathbf{U}$ given by Eq. (42).

Figure 30 shows the BLUE-combined dressed cross section results for the three invariant mass regions: the $\omega(782)$ region $(0.7 - 0.9$ GeV), the $\phi(1020)$ region $(0.9 - 1.05$ GeV), and the ω'/ω'' region $(1.05 - 2.0$ GeV).



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω'/ω'' region

Figure 30: BLUE-combined dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$ results for the three invariant mass regions. The error bars represent the statistical uncertainties propagated from the covariance matrices. The combination significantly improves the statistical precision compared to any single data-taking period.

For completeness, Fig. 31 presents the combined results across all three regions on a logarithmic scale, illustrating the full spectrum from 0.7 to 2.0 GeV in a single view.

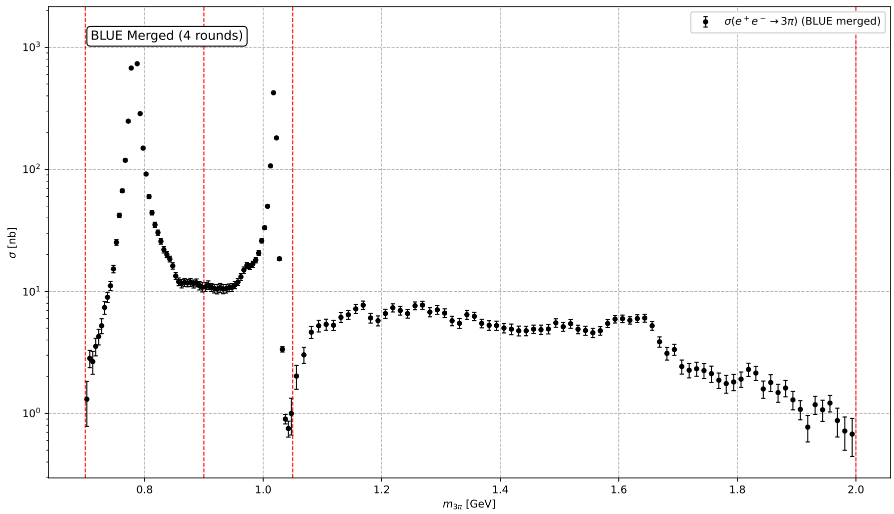


Figure 31: BLUE-combined dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$ across all three invariant mass regions, shown on a logarithmic scale.

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APPENDIX